IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 5 August 2025, accepted 22 August 2025, date of publication 28 August 2025, date of current version 8 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3603868

RESEARCH ARTICLE

Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions

ZAFRIR AVRAHAMI[1,2], MOTI ZWILLING[1,2], AND CHEN HAJAJ[2,3], (Member, IEEE)
[1]Department of Economics and Business Administration, Ariel University, Ariel 40700, Israel
[2]Ariel Cyber Innovation Center, Ariel University, Ariel 40700, Israel
[3]Department of Industrial Engineering and Management, Ariel University, Ariel 40700, Israel

Corresponding author: Zafrir Avrahami (Zafi.av@gmail.com)

ABSTRACT The growing complexity of the digital environment has increased the need for proactive and intelligence-based approaches to cybersecurity. This study examines the role of open-source intelligence (OSINT) as a strategic tool in proactive cybersecurity operations. Drawing on a wide range of peer-reviewed literature and professional sources, it reviews definitions, operational processes, areas of application, benefits, and challenges associated with OSINT. The analysis highlights OSINT's contribution to situational awareness, early threat detection, and cyber threat intelligence (CTI) capabilities. By using publicly accessible data from the internet and social platforms, organizations can strengthen their defensive posture against diverse cyber threats. The study outlines workflows for collecting and analyzing OSINT, with attention to its integration into organizational intelligence frameworks and cybersecurity strategies. Examples from both the public and private sectors demonstrate how OSINT supports decision-making, incident response, and preparedness for emerging threats. The review also considers OSINT's main advantages, including cost effectiveness, accessibility, and real-time relevance, alongside its main challenges such as data reliability, legal concerns, and information overload. The findings suggest that, despite these limitations, OSINT has significant potential to enhance proactive cybersecurity measures, support compliance with standards, assist law enforcement, prevent terrorism, and contribute to business decision-making. The paper concludes with a call for further research on integration with advanced technologies, real-time data analysis, and effective intelligence collaboration.

INDEX TERMS Open-source intelligence, OSINT, cyber threat intelligence, CTI, intelligence cycle, proactive cybersecurity.

LIST OF ABBREVIATION

AI	Artificial Intelligence.
API	Application Programming Interface.
APT	Advanced Persistent Threat.
CERT	Computer Emergency Response Team.
CJCS	Chairman of the Joint Chiefs of Staff.
CTI	Cyber Threat Intelligence.
DNS	Domain Name System.
GDPR	General Data Protection Regulation.
HIPAA	The Health Insurance Portability and Accountability Act.
HUMINT	Human Intelligence.
IoC	Indicators of Compromise.
ISACs	Information Sharing and Analysis Centers.
ISAOs	Information Sharing and Analysis Organizations.
IR	Incident Response.
ML	Machine Learning.
MSSPs	Managed Security Service Providers.
NAC	Network Access Control.
NIST	National Institute of Standards and Technology.
NLP	Natural Language Processing.
OE	Operational Environment.
OPSEC	Operations Security.
OSINT	Open-Source Intelligence.
PII	Personally Identifiable Information.

The associate editor coordinating the review of this manuscript and approving it for publication was Mohamad Afendee Mohamed.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

PT	Penetration Testing.
SIEM	Security Information and Event Management.
SOAR	Security Orchestration, Automation, and Response.
SOC	Security Operations Center.
STIX	Structured Threat Information Expression.
TECHINT	Technical Intelligence.
TI	Threat Intelligence.
TTPs	Tactics, techniques, and procedures.
VISINT	Visual Intelligence.
WAF	Web Application Firewall.
WWW	World Wide Web.

I. INTRODUCTION

We live in an era dominated by digital devices. These technologies have transformed daily life but have also exposed society to significant and unprecedented cyber threats [1]. As the digital domain continues to expand, cyber attackers adapt their tactics and exploit vulnerabilities for purposes such as financial gain or political advantage. In today's dynamic and complex cyber environment, proactive intelligence-based defense measures are critical. Open-source intelligence (OSINT) and cyber threat intelligence (CTI) serve as essential tools for strengthening organizational defenses and ensuring preparedness against future cyber threats.

Today, many analysts across intelligence disciplines begin investigations by collecting open-source intelligence (OSINT), and only afterward incorporate additional sources [2], [3]. Intelligence services currently obtain more than 80% of their intelligence from open sources [4]. Cyber threat intelligence (CTI) includes various intelligence domains used to defend against cyberattacks, such as OSINT, human intelligence (HUMINT), and visual intelligence (VISINT). This review highlights the significant role of OSINT in CTI. Integrating OSINT supports organizational cybersecurity in areas such as risk assessment, incident response, detection of advanced persistent threats (APT), and early warning of cyber threats. It also identifies additional applications for OSINT, including support for law enforcement, counterterrorism, business intelligence, and deepfake detection.

This study is valuable for several reasons: (1) The cyber threat landscape is constantly evolving, with new threats, attack techniques, and security vulnerabilities emerging regularly. This study explains how to effectively leverage OSINT data to **keep pace with these changes**; (2) OSINT offers a wealth of publicly available information that can enhance threat intelligence. This study demonstrates how OSINT sources contribute to a more comprehensive understanding of cyber threats and support more **informed decision-making** in organizational cybersecurity planning; (3) OSINT systems collect and analyze large volumes of data, so it is essential to identify the most relevant and reliable information. Iterative processes allow for more accurate intelligence collection, helping cybersecurity teams **allocate resources more efficiently** based on organizational needs; (4) OSINT helps identify threat actors and their motivations, supporting more accurate and reliable **threat analysis**; and (5) as organizations increasingly integrate OSINT into their cybersecurity strategies, this study presents **best practices and guidelines** for the effective use of data derived from OSINT.

This study provides an overview of OSINT in the context of cybersecurity, focusing on its definitions, workflows, applications, compliance with cybersecurity standards, benefits, and challenges. It synthesizes insights from existing academic literature and aims to identify strategic approaches for integrating OSINT into cybersecurity operations while suggesting directions for future research. Additionally, this study seeks to contribute to a deeper understanding of how OSINT can be leveraged to enhance organizational cyber resilience and threat preparedness. By framing OSINT as a core enabler of proactive cybersecurity, this study establishes a conceptual foundation for future empirical and theoretical studies.

This study argues that OSINT plays a key role in proactive cybersecurity by enabling earlier detection of threats, supporting faster response times, and improving decision-making through access to diverse and up-to-date information. OSINT helps organizations identify emerging risks, track threat actors, and adapt to evolving cyber challenges before they cause damage. This manuscript presents a structured overview of OSINT processes, tools, applications, and challenges, and shows how each element contributes to anticipating, identifying, and responding to cybersecurity threats in a proactive manner.

II. BACKGROUND AND LITERATURE REVIEW

A. OPEN-SOURCE INTELLIGENCE (OSINT): DEFINITION

Open-source intelligence (OSINT) refers to the collection, processing, and correlation of publicly available information from open sources such as mass media, social networks, forums and blogs, public government data, and commercial publications or datasets. OSINT systems gather data from multiple sources using advanced collection and analysis techniques, enabling a broader intelligence picture of the target. The information amassed feeds back into the collection process to refine and move closer to the final objective [2], [3], [5].

The U.S. Department of Defense defines open-source intelligence (OSINT) as intelligence derived from publicly available information that is collected, used, and distributed in a timely manner to the appropriate audience in response to a specific intelligence requirement [6]. OSINT is produced through the systematic collection, processing, and analysis of relevant public information to meet this requirement. The two key terms in OSINT are "open sources" and "publicly available information." Open sources refer to information provided by any person or group without an expectation of

privacy. Both the content and its links are not protected from public exposure and are accessible to the general public. Publicly available information includes data, facts, instructions, or other material published or broadcast for public use and legally viewable by anyone [3], [7], [8].

A significant milestone in the evolution of OSINT was the widespread use of the internet in the early 2000s and the bursting of the dot-com bubble, which led to an exponential increase in the volume of data available online. At the same time, social media platforms emerged, allowing the creation of personal profiles and interest groups [4]. In 2005, the internet underwent a major shift, as most online content moved to dynamic web-pages. One of the key impacts was the transition from Web 1.0 (one-way flow of information from source to consumer) to Web 2.0 era (where information consumers actively participate in content creation, such as posting comments, information, and multimedia). This transition exponentially increased the volume of user-generated data available for sharing and interaction. Social networks such as Facebook.com and YouTube.com were launched in 2005, and Twitter was founded in early 2006 [2].

The term OSINT became associated with national security and military agencies [9], [10]. Initially used as a tactic for military intelligence, its application later expanded to business intelligence, politics, and law enforcement. For example, corporations use OSINT strategically, academics use it for research, and law enforcement agencies use it to counter national security threats and terrorism [11]. Until the 1990s, most advanced OSINT technologies flowed from the security domain to the business sector. However, after 2000, this trend reversed. The academic and business sectors significantly contributed to technological advancements. One example is the "Hacking for Defense" [12] project, which involves the U.S. Department of Defense (DOD), U.S. Intelligence Community (IC–US), 31 universities, and 66 companies [2], [13].

People are connected to the internet all day, documenting their activities and sharing what happens around them. As of February 2025, there were approximately 5.56 billion internet users worldwide, representing 67.9% of the global population. Of these, approximately 5.24 billion were social media users, accounting for 63.9% of the global population [14]. To illustrate the volume of data uploaded online, Statista reports the amount of content shared in a single minute on the global internet. For example, in 2024, 251.1 million emails were sent per minute, 138.9 million Reels were played per minute on Facebook and Instagram, and 5.9 million Google searches were performed per minute [15].

Today, open-source intelligence (OSINT) is widely adopted by governments and intelligence services to conduct investigations and combat cybercrimes [16]. OSINT is also of interest to companies and individuals for purposes such as targeted information retrieval, social media analysis, and identifying connections between people and civil organizations [17]. In recent years, anyone using an internet search engine is essentially using OSINT technology to find publicly available online information. These private users "collect" OSINT data for everyday needs such as education, work, research, and personal matters. For example, as of January 2025, Google remained the most popular search engine globally, holding a dominant market share of 89.62%, followed by Bing with 4.04%, Yandex with 1.63%, and Yahoo with 1.34% across all devices [18].

B. CYBER THREAT INTELLIGENCE (CTI): DEFINITION

Cyber threat intelligence is sometimes referred to simply as threat intelligence. **Intelligence is defined** as information that can be acted upon to alter outcomes [19]. **A cyber threat** is any circumstance or event with the potential to negatively impact organizational operations (including mission, functions, image, or reputation), assets, or individuals through the compromise or alteration of information systems, unauthorized access, destruction, exposure, and/or denial of service [20]. **Threat information** refers to any data related to a threat that can help an organization detect malicious activity and defend against it. Cyber-threat information may include indicators of compromise, tactics, techniques, and procedures (TTPs), security alerts, threat intelligence reports, and defensive tool configurations [21].

Cyber threat intelligence (CTI) is evidence-based knowledge that includes relationships, mechanisms, indicators, implications, and actionable recommendations regarding an existing or emerging threat or hazard to organizational assets. This intelligence can be used to inform decisions about the organization's response to the threat [?]db@bib:22. CTI includes technical information such as attack vectors, threat actors, victims, methods of operation, and affected organizations. This information is delivered through CTI feeds using standard formats, such as MAEC, STIX, TAXII, and CYBOX [23]. Additionally, cybersecurity communities have established CTI platforms for threat sharing, such as MITRE, MISP, Microsoft, and UK/US CERT (Computer Emergency Response Team) [24].

Effective cyber threat intelligence relies on information that is [25], Fig. 1 shows the key characteristics of effective cyber threat intelligence:

- **Timely:** Intelligence must be delivered quickly and without unnecessary delay, ideally in real time, and should include information on threat actors.
- **Relevant:** The collected intelligence must be applicable to the organization's operational environment, address potential threats the organization may face, and provide insights into the risk level of the specific threat.
- **Accurate:** Intelligence must be high-quality, complete, and consistent. Ambiguity that could lead to incorrect actions, inappropriate responses, or a false sense of security must be avoided.
- **Specific:** Intelligence must contain concrete and meaningful details about threat actors and explain how attacks or threats could impact the organization's systems.
- **Actionable:** Intelligence should include recommended actions that can be taken in response to the threat to

prevent harm to the organization or reduce the impact of an attack.

Cyber threat intelligence can be obtained from internal or external sources in the organization. **Internal sources include** internal network event data, DNS log files, firewall logs, alerts, incident response (IR) reports from previous events, retained samples of malware previously used in attacks, and captured relevant network packets [26]. **External sources include** open-source intelligence, structured and unstructured reports, email exchanges from sharing groups, cyber intelligence team blogs, government blogs, and any publicly available information that provides indicators relevant to the organization [27].

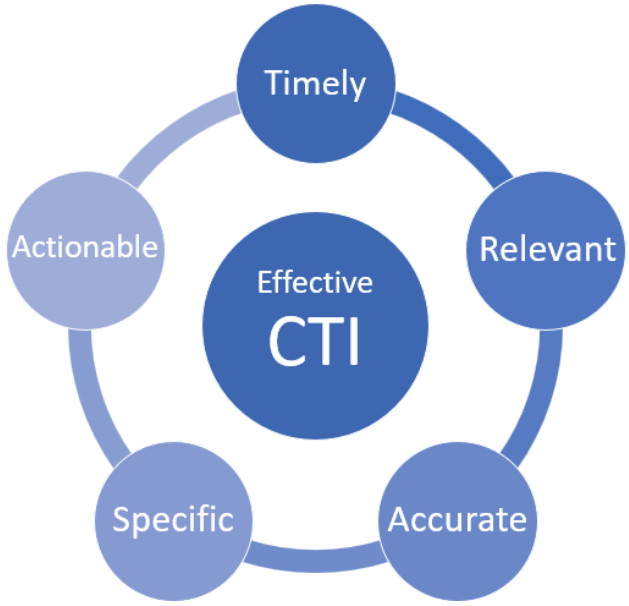

FIGURE 1. Effective cyber threat intelligence.

C. THREAT INTELLIGENCE: FEATURES

Threat intelligence refers to any information that can inform decisions. Fig. 2 [19] shows a model that divides threat intelligence into four categories based on consumption:

- **Strategic threat intelligence** provides an overview of the threat landscape and the impact of attack trends on executives' business and financial decision-making. This type of report should be brief, concise, and focused [28].
- **Operational threat intelligence** delivers information about the details of an incoming attack, the identity and capabilities of the attacker, and the likelihood of an attack. This is intended for cybersecurity managers and incident-response team leaders [29].
- **Tactical threat intelligence** offers more detailed information, including the adversary's tools, methodologies, technical capabilities, and tactics, techniques, and procedures (TTPs). These data were collected from malware reports, incident reports, attack group reports, and human intelligence [30].
- **Technical threat intelligence** provides technical details about an attacker's capabilities, such as the tools used, command and control (C2) channels, and attack infrastructure. This information is aimed at frontline defense teams and SOC analysts [31].

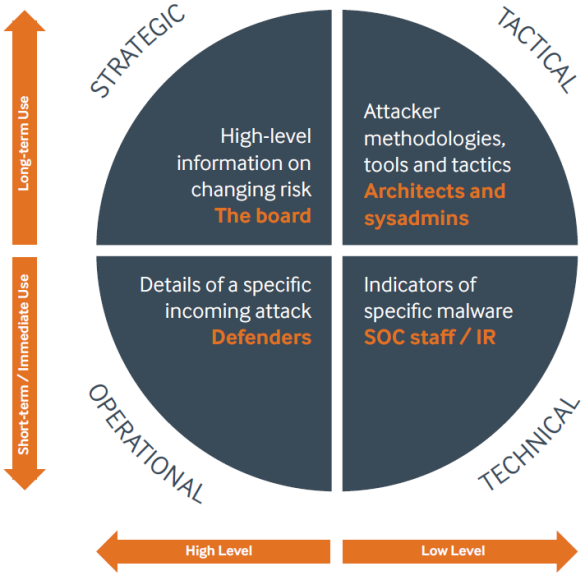

FIGURE 2. Subtypes of threat intelligence.

Cyber threat attribution refers to the identification of an individual or organization responsible for an attack, based on different attacker profiles and characteristics [32]. As shown in Fig. 3. There are three levels of attribution: **the first** involves identifying the tactics, techniques, and procedures (TTPs) used by the attacker; **the second** includes information about the country, motivations, and objectives behind the attack; and **the third**, most critical level, containing specific details about the attackers themselves. For example, advanced persistent threat (APT) attacks are typically state-sponsored, making attribution particularly challenging. A key task in cyber threat attribution is determining the current stage of the attack based on MITRE's cyber kill chain framework [33].

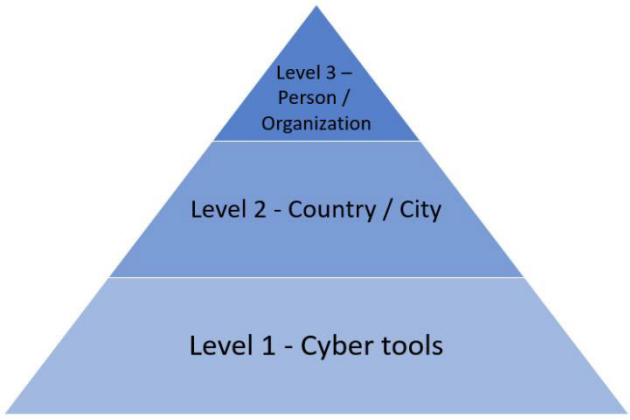

FIGURE 3. Cyber threat attribution levels.

D. PROACTIVE CYBER PROTECTION

Cybersecurity strategies must also advance to keep pace with the evolution of cyber threats. In early 2024, the National Institute of Standards and Technology (NIST) released version 2.0 of its Cybersecurity Framework. This updated version helps organizations establish and maintain a proactive cybersecurity strategy tailored to their specific risk environment. Given the current threat landscape, organizations are encouraged to shift from a reactive approach to a proactive defense posture [34]. The term proactive is defined as taking action to make changes yourself rather than reacting to things that happen [35].

A proactive cybersecurity strategy focuses on anticipating future problems, needs, or changes, and implementing continuous actions and processes to identify, mitigate, or prevent threats. It should include CTI, regular penetration testing, vulnerability scanning, monitoring, cybersecurity awareness training, and behavior-based analysis. Proactive defense can help organizations manage risks, stop threats before they occur, respond effectively, recover quickly, and significantly reduce costs [36]. In today's dynamic and complex cyber environment, proactive intelligence-driven defense is critical. This is where CTI and OSINT become essential tools for strengthening defenses and improving readiness for future cyber threats.

III. OSINT: WORK PROCESSES

Proactive cybersecurity requires structured processes that transform open data into a timely and actionable intelligence. This chapter describes the intelligence cycle, production processes, data handling, and core tools and techniques used in OSINT workflows. Each component is essential for enabling early detection, an efficient response, and information sharing to support preventive security efforts.

A. INTELLIGENCE CYCLE

The intelligence cycle, also known as the lifecycle, is the process of transforming raw data into finished intelligence for decision-makers to support action [37], [38]. In the U.S. military, it is referred to as the "intelligence process" and consists of six phases. In 2013, General Martin Dempsey, Chairman of the Joint Chiefs of Staff (CJCS), issued updated guidance for conducting joint and multinational intelligence operations. This directive serves as the intelligence doctrine for coordinating the U.S. Army with other government departments and agencies during operations. Its purpose is to structure the force, execute the mission, and ensure unity of effort toward achieving the overarching goals. The directive outlines six interconnected categories of intelligence activities designed to provide relevant and timely intelligence to commanders and national-level decision-makers. The intelligence cycle shown in Fig. 4 is based on [39], which has been adopted as a standard operating model by security organizations worldwide, including in the UK and Australia, and is still in use today [19], [40], [41].

FIGURE 4. The intelligence process.

1. **Planning and direction:** This phase involves defining areas of interest, preparing a collection plan, setting priorities, developing an appropriate intelligence architecture, and submitting requests to data collection agencies. A subtopic of this phase is Intelligence Requirement, which refers to any general or specific topic that requires information collection or intelligence production [39]. These intelligence requirements must align with and support the goals and activities of the organization or client. This stage is critical because it initiates the intelligence cycle. Additionally, it identifies the best source of information to serve as the basis for data collection in the next phase of the cycle [40].
2. **Data collection:** In this phase, relevant data is retrieved from publicly available open sources based on the target objective. The internet serves as a primary source of information because of the vast amount of accessible information. The goal is to extract specific facts from a massive pool of open and accessible resources such as search engines, social networks, and blogs. To begin the search, at least one data point about the target is required, for example, an email address, username, real name, location, or IP address. Using this initial piece of information, the investigator applies OSINT techniques to generate additional data. The data obtained through a single technique serve as input for generating additional data with other techniques. From this stage onward, the entire intelligence creation process is initiated [10], [42].
3. **Processing and exploitation:** This phase, also referred to as data enrichment, involves transforming the collected raw data into understandable and valuable information. On their own, the data are not useful and must be interpreted to derive initial facts through a

preliminary analysis. Continuous iterations and analysis using various OSINT techniques are required to generate actionable insights, such as information retrieval, organization, classification, extraction, translation, and distinguishing signals from noise [10], [42].

4. **Analysis and production:** This phase involves knowledge extraction and inference. The information generated in the previous phase is used as the input for advanced inference algorithms, such as pattern recognition, profiling behavior, value prediction, and event correlation [42].
5. **Dissemination and integration:** In this phase, intelligence is delivered to the consumer and put to use. The method of dissemination is determined by the client's needs and the criticality of intelligence. Cyber intelligence companies must initiate and maintain close contact with clients, ensure the timely delivery of intelligence products, and confirm that these products meet requirements. Intelligence personnel are responsible for ongoing support even after delivery, aiding in decision-making and responding to follow-up questions, emerging issues, or the need to relate products to a broader intelligence picture. Integration is an ongoing dialogue between clients and intelligence professionals. How intelligence is used depends on the client, whereas the role of intelligence personnel is to provide the best possible intelligence [39], [40].
6. **Evaluation and feedback:** Evaluation and feedback occur continuously throughout all the stages of the intelligence cycle. Intelligence personnel at all levels must assess the execution of intelligence tasks and examine their outcomes. This phase requires ongoing dialogue between all intelligence personnel involved in production and intelligence consumers. The goal of evaluation and feedback is to identify issues as early as possible to minimize information gaps and mitigate capability shortfalls [39], [40].

Iteration is a repeated action performed during the process of solving a problem. The intelligence cycle is an iterative process in which data is continuously fed into the system to produce a sequence of ongoing results. The process begins with data collection, followed by data enrichment and knowledge inference, and then loops back to the initial stage, repeating in a cyclical manner. The feedback from the inferred knowledge can be compared with the initial parameters of the data collection phase [43].

Validated OSINT (OSINT-V) refers to OSINT that has a high level of certainty and reliability. This information must be verified and validated through a trusted and reputable source. Validation is essential because skilled cyber attackers also analyze OSINT and deliberately generate and distribute false information across various websites on the internet [41].

Operational Environment (OE) refers to the context in which raw data is collected from across the internet. The strategy for data collection varies depending on the specific objectives of the search. OSINT technologies and systems enable access to large volumes of internet-based **data** almost in real time. This data is raw material with limited immediate value. It must be processed into a structured form to become **information** that can be used. When this information is connected to other information collected in the operational environment or from previous investigations, it can lead to a new understanding, defined as **intelligence**. Intelligence can also be produced by analysts who compare data across sources and databases and draw conclusions. The relationship between data, information, and intelligence is illustrated graphically in Fig. 5. Intelligence helps anticipate or forecast future conditions and supports decision-making by highlighting key findings. It is important to remember that: (1) intelligence is not an exact science; (2) impactful intelligence depends not only on the provider but also on understanding the specific needs of the end-user; and (3) since the internet is a dynamic environment, intelligence generation must be an iterative and ongoing process [39], [40], [44].

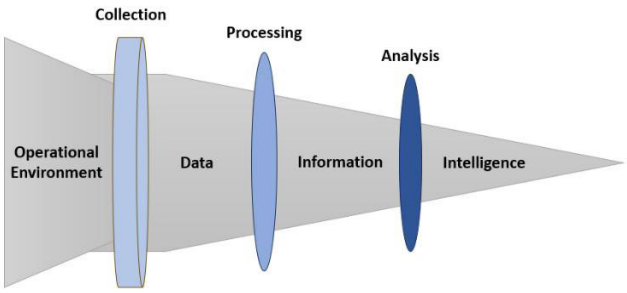

FIGURE 5. The relationship between data, information and intelligence.

Illustrative Example – From OSINT Data to Actionable Intelligence: An analyst monitoring open sources detects a suspicious domain name such as *secure-login-update.com* through a free domain monitoring tool. At first glance, this domain seems legitimate, but additional OSINT tools reveal that it was registered recently and is hosted on a server with a history of distributing malware. By checking threat intelligence databases and examining technical indicators, the analyst finds that the same server has been linked to phishing emails targeting employees in the finance sector. Based on this evidence, the analyst concludes that this domain is likely part of an active phishing campaign. A clear recommendation is made to block access to the domain within the organization's firewall and update the email filters to detect related phishing attempts. The diagram in Fig. 6 shows this simple process that shows how raw OSINT data can be turned into a useful and actionable security decision.

B. DATA COLLECTION AND INTELLIGENCE PRODUCTION

OSINT draws upon a wide range of data sources, both digital and physical. These sources form the foundation of any intelligence-gathering effort and determine the scope and quality of insights that can be extracted. Table 1 presents a classification of common online and offline OSINT sources,

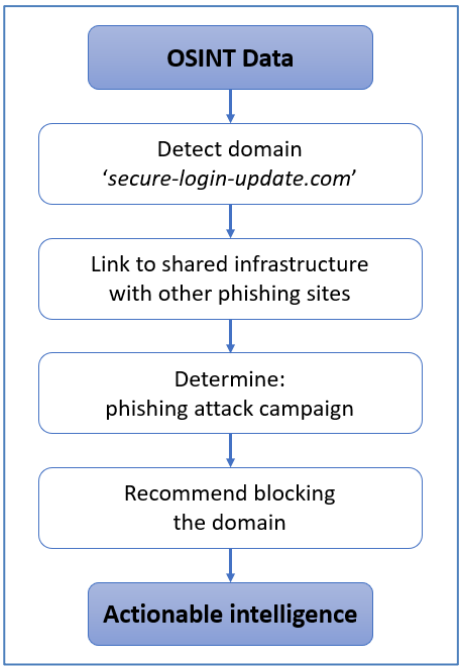

FIGURE 6. From OSINT data to actionable intelligence (diagram).

TABLE 1. Sources of online and offline OSINT.

Online OSINT	Offline OSINT
Social media	Newspapers
Websites	Television
Deep web	Radio
Dark web	Public Records
Online databases	Libraries: Books
Online datasets	Court records / legal filings
Academic research	Conferences & trade publications
Government websites	
Search engines	
Domain name system	
Online Forums	
Geolocation data	
Code repositories (e.g., GitHub)	
Multimedia platforms (e.g., YouTube, TikTok)	
E-commerce platforms (e.g., marketplaces)	

highlighting the diversity of channels from which actionable information can be collected.

The internet is a dynamic information environment composed of virtual websites and both static and dynamic target sites containing a mix of old and new content [7]. It can be divided into three networks:

- **Surface web:** This is the part of the internet familiar to most users. It includes the World Wide Web (WWW), indexed by standard search engines. These search engines cover only about a quarter of the information available on the internet [7], [45].
- **Deep web:** This includes all password-protected or encrypted websites, typically without special identifying information (anonymization). It also includes government sites, closed-access academic articles, and databases that may be unknowingly accessible via the internet. Social media profiles set to private are also part of this network, which is not part of traditional WWW. The content in the deep web is estimated to be more than twice the volume of the surface web, and is not indexed by standard search engines [7], [45].
- **Dark web:** This layer offers user anonymity and privacy, which allows criminals to browse, search, and post data for illegitimate purposes while concealing their identities. As a result, the dark web is an ideal source for applying OSINT in the fight against cybercrime and organized crime. On the other hand, tracking and de-anonymizing criminals in this environment poses significant challenges for OSINT experts [45]. Collecting information from this network requires specific tools [46], [47].

OSINT tools and techniques for intelligence collection include:

- **Web crawlers built with programming structures:** A web crawler is a programmed tool that scans web pages by following hyperlinks found within them [10], [42].
- **Web scraping:** This technique extracts data from websites using dedicated protocols. The software accesses web pages automatically through bots or crawlers, collects specific data (a dataset), stores it, and uses it later [13].
- **Sock puppet accounts:** Also known as research accounts, these are fake online identities used by OSINT investigators to hide and protect their identity. These accounts enable access to information that requires a subscription or user account, especially on platforms such as social media, where content may only be available to logged-in users or for isolating research activities [48].
- **Third-party APIs:** Application programming interfaces, or APIs, allow software applications to communicate with one another. Social media sites, search engines, and news websites provide APIs that enable software to query and retrieve data [10], [42].
- **Automated tools:** These commercial tools provide data through an easy-to-use interactive interface. They aggregate data from multiple sources and can enrich it, transforming it into high-value information. They also offer filters for including or excluding specific data sources and data types [10].

Processing stage:

During the processing stage, the collected information must be normalized, aggregated, indexed, and stored so that it can be easily identified and used. Normalization refers to converting and standardizing all collected information into a uniform format. For example, all data should be in digital text form, in the same language, and all images should be in a specific digital format [49]. Most information processing tasks are performed using data processing tools and artificial intelligence. Examples of these tools include information

filtering, automatic classification, machine translation, and automatic speech recognition. Since these tools have limitations, some tasks must still be performed manually or semi-automatically by intelligence analysts [38]. Knowledge organization systems are designed to organize documents through interpretation, semantic analysis, synonym control, and the creation of explicit and associative semantic links. These systems support sense-making by clarifying concepts and the relationships between them in the context of a given problem or task, which is an essential part of intelligence analysis and interpretation [38], [50].

During the **data enrichment stage**, the goal is to transform collected data into meaningful information by identifying and separating relevant content. This process involves two main steps: **Step 1: Extraction of relevant data** – Textual information is extracted from raw text using Natural Language Processing (NLP) techniques based on machine learning and deep learning models. Text recognition can be either (1) *Assisted recognition*, which uses pre-labeled data, past observations, or predefined keywords, or (2) *Unassisted recognition*, where NLP algorithms learn patterns from the text without using pre-labeled training data. Examples include: (i) *Topic modeling*, a machine learning technique applied to large, heterogeneous text corpora to identify themes based on frequently occurring words. This model groups sentences according to their topic. (ii) *Latent Dirichlet Allocation (LDA)*, a probabilistic model that identifies word dependencies without relying on semantic (syntactic) similarities [10], [50].

Step 2: Feature extraction – Once relevant data is identified, specific attributes related to the topic can be extracted. One widely used NLP tool in this step is *Named Entity Recognition (NER)*, which identifies entities in unstructured text and classifies them into predefined categories such as person names, locations, dates, and times. Additional domain-specific entities may also be relevant. For instance, in the context of a cyberattack, entities might include the attackers, type of attack, tools used, and the targeted organization [10], [51]. Advanced data mining and NLP techniques enable the extraction of meaningful patterns from unstructured data with complex relationships, which are essential for intelligence generation [42]. NLP algorithms can also analyze social media posts, forums, blogs, and other online content to detect sentiments, intent, or early signs of malicious activity, providing a valuable context for emerging threats.

Personally Identifiable Information (PII) refers to any information that relates to a specific individual and can be used to reveal their identity or distinguish them from others. Examples of PII include name, address, email, phone number, date of birth, passport number, fingerprints, driver's license number, credit card number, and Social Security number [49].

Tools and techniques:

OSINT operations rely on a wide array of tools designed for specific data acquisition and analysis tasks. These tools range from device search engines to data aggregation platforms and dark web scanners. Table 2 presents a selection of ten widely used OSINT tools and their primary functions, reflecting the diversity of capabilities available to analysts. For a comprehensive list of OSINT tools, sources and frameworks please see the OSINT framework [52].

TABLE 2. Selected automated OSINT tools and their primary uses.

Tool	Primary use/function
Maltego	Graphical link analysis and data correlation across multiple sources
Shodan	Internet-connected device discovery and network intelligence
SpiderFoot	Automated reconnaissance across multiple data sources
TheHarvester	Collection of emails, domains, subdomains, and usernames
Recon-ng	Modular web reconnaissance framework
Intelligence X	Search engine for leaked data, dark web, and historical web content
BuiltWith	Website technology profiling (servers, CMS, frameworks)
Whois	Domain ownership and registration lookup
OnionScan	Scanning and analysis of Tor hidden services (dark web OSINT)
ExploitDB	Database of known software vulnerabilities and exploits

In addition to specialized tools, OSINT relies on a set of practical techniques that guide how information is collected and analyzed. These methods allow analysts to extract, verify, and contextualize data from both structured and unstructured sources. Table 3 presents five widely used OSINT techniques commonly applied in cyber threat intelligence, investigations, and digital forensics.

TABLE 3. Selected OSINT techniques and their use.

Technique	Primary use/function
DNS Lookup	Mapping domain and subdomain ownership and infrastructure
Reverse Image Search	Identifying visually similar or identical images across the web
Social Media Investigation	Collecting information about individuals, groups, and organizations
Google Dorks	Advanced search queries to uncover hidden or sensitive data
Scraping	Automated collection of publicly available web data

OSINT Framework: This open-source project developed and maintained by the OSINT community and volunteers. The OSINT Framework is available online as a free resource for the OSINT community. The official website is (osint-framework.com). Fig. 7 shows, a screenshot of the work on the site. This framework serves as an important starting point for OSINT investigations. It is a comprehensive collection of open-source intelligence tools, resources, and techniques organized within a structured framework, serving as a central platform for practitioners and researchers. This framework includes data sources, techniques, tools, and resources for locating individuals, usernames, and email addresses [42], [52], [53].

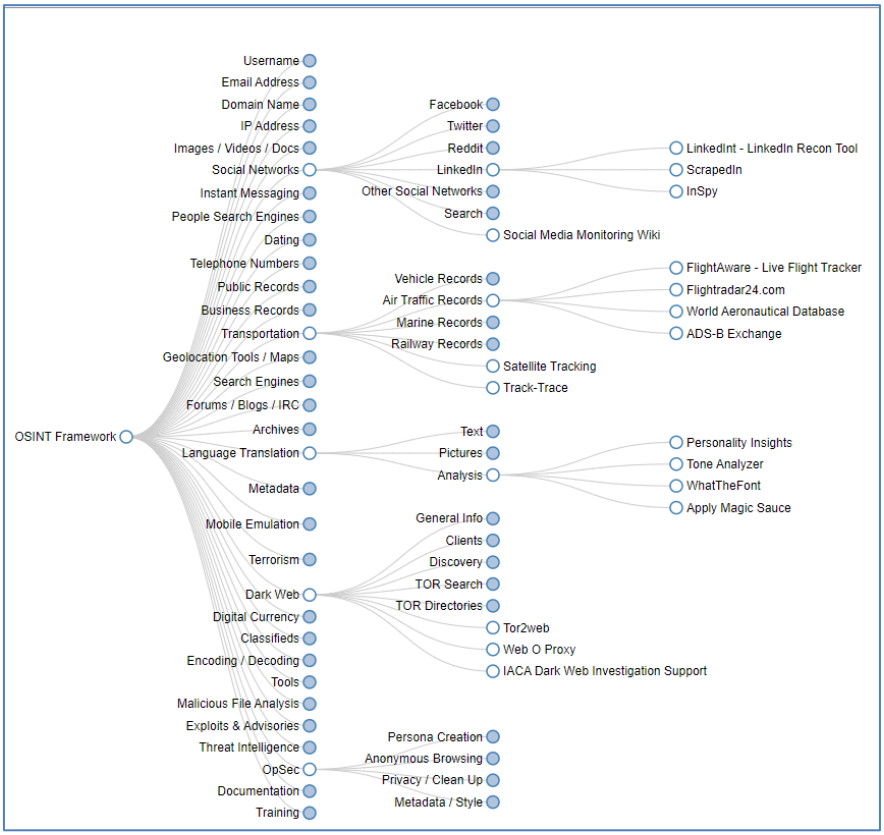

FIGURE 7. OSINT framework site.

- **STIX standard:** Structured Threat Information expression (STIX) is a standardized language and framework for representing, exchanging, and sharing cyber threat intelligence and OSINT awareness [23], [54], [55].
- **Indicators of compromise (IoC):** Indicators of compromise (IoCs) are findings or evidence that suggest a security breach or attack has occurred within computer networks, systems, or infrastructures. IoCs are used in cybersecurity investigations to identify and detect malicious activities, assess the scope of a security event, and implement appropriate response measures [56], [57].
- **Operations Security (OpSec):** This is a strategic operational security approach that involves a set of principles, practices, and measures implemented to protect sensitive or classified information [7], [39], [58].

C. INTELLIGENCE SHARING

In the traditional intelligence domain, information sharing is restricted (compartmentalization) to relevant individuals in to minimize exposure and risk. In the field of cyber threat intelligence, however, the guiding principle is the "need to share." Sharing occurs through the creation of communities and collaborative relationships, allowing all participants to benefit from cyber threat intelligence. An organization can be harmed by an attack on a competitor, as attackers may reuse the same information and techniques against others in the same sector. Many cyberattacks are not isolated incidents but are aimed at multiple organizations, often within the same industry, meaning that analyzing and understanding these attacks can benefit all related entities. When entire communities are targeted, they must collectively defend themselves. The goal is to raise the bar and continuously increase the cost for attackers [19].

1) PRECAUTIONS IN INFORMATION SHARING

Effective sharing requires trust, because the information being shared may be sensitive. For example, if an organization admits that it has been attacked, that information could reach the attackers, who might then change their methods. Closed and trusted groups can support deeper sharing, such as through government portals, private forums, mailing lists, and reliable personal connections with intelligence personnel in peer organizations.

2) WHEN NOT TO SHARE INFORMATION

In certain sectors, even a rumor about a threat can influence purchasing decisions or market evaluations. Disclosing vulnerability to competitors is not natural, and it is important not to reveal information that could have a negative impact on business. Trust must be built with specific individuals

at competing companies to prevent non-trusted parties from misusing shared information. In some industries, organizations may feel more secure using a trusted third party, such as a Computer Emergency Response Team (CERT), to anonymize and distribute the information, enabling community benefit while minimizing reputational risk [19].

The U.S. National Institute of Standards and Technology (NIST) has published guidelines for establishing information-sharing relationships regarding cyber threats. This guidance helps organizations in setting information-sharing objectives, identifying sources of cyber threat intelligence, creating rules for the publication and distribution of threat information, engaging with intelligence-sharing communities, effectively utilizing threat information within the organization, and supporting cybersecurity methodologies. Organizations can leverage collective knowledge to gain a more comprehensive understanding of potential threats and make informed decisions in their organizational risk management strategies [21]. The leading global standards for intelligence sharing include: STIX, the U.S. standard for automated threat information sharing; the Malware Information Sharing Platform (MISP), the European Union's standard; and private entities such as MITRE Corporation, which has also developed frameworks for information sharing [40].

IV. OSINT: APPLICATIONS

Translating OSINT into proactive defense depends on how it is applied in real-world settings. This chapter presents practical applications in cybersecurity, explores differences in how the public and private sectors use OSINT, and provides a case study from the United States. These examples illustrate how OSINT can help organizations anticipate and respond to threats before they escalate.

A. APPLICATIONS IN CYBERSECURITY

Cyber threat hunting is a proactive and recurring process of actively searching for security threats, indicators of compromise (IoCs), and malicious activities targeting an organization or individual. Driven by expert analysts, it involves identifying tactics, techniques, and procedures (TTPs) within the organization's network. Instead of relying on alerts triggered by defensive tools, this approach actively seeks threats. It is an iterative process involving continuous analysis of network data to detect IoCs that may indicate malicious behavior. Data collection occurs both within the organization and from open sources (OSINT) to provide a broad view of potential threats and activities that could affect the organization. The process includes threat analysis, IoC identification, reporting, and response, aiming for early detection and mitigation of sophisticated, unknown threats that may have bypassed traditional defenses [44], [59], [60].

Addressing Advanced Persistent Threats (APT): Security information and event management (SIEM) systems provide deep protection across all layers of organizational assets. These systems can compile and coordinate large volumes of data transmitted from network sensors such as firewalls, Web Application Firewall (WAF), and Network Access Control (NAC). They monitor network activity and significantly enhance threat detection capabilities. However, APTs emerged as a type of attack aimed at data theft rather than damaging organizational networks. These attacks are characterized by four main features: (1) specific and clear targets; (2) high-level organization and substantial resources; (3) extended attack duration; and (4) a high degree of stealth. SIEM systems have proven to be insufficient against these advanced and sophisticated threats. In contrast, cyber threat intelligence (CTI), through access to intelligence updates from open sources (OSINT), has become a key asset in uncovering APTs by enabling early detection and anticipation of future attacks, thereby significantly enhancing the organization's defensive posture [56].

Integrating OSINT into cyberattack investigations: Due to the growing online exposure of companies and organizations, they are increasingly required to invest significant resources in cybersecurity to protect their assets from attackers. Therefore, it is essential to implement effective detection and response mechanisms, as well as threat and incident management for organizational information systems. Incorporating CTI within the organization is necessary for extracting and analyzing traces, patterns, and conclusions from incidents, so as to complement traditional technical defense mechanisms with up-to-date information that significantly enhances infrastructure protection, vulnerability management, and response effectiveness. The attack profile can be completed with valuable information derived from cyberattack traces and cross-referenced with open-source incident and investigation databases to identify root causes, understand attacker motivation, and even construct attacker profiles [42].

Integrating OSINT into penetration testing (PT): During information security audits, the first task of a penetration testing (PT) expert is to search for all publicly available information about the organization using open sources. Numerous technologies and resources are available for finding documents, databases, personal data, and other organizational assets. A study by [61] emphasized the need to automate this process. Typically, the tester is only given the organization's name in advance, and through OSINT sources, a large amount of information can be obtained, such as the organization's area of activity, phone numbers, address, email, and full names of executives and owners.

Collaboration with an incident response (IR) team: Threat intelligence can be used in various ways to help an organization defend against incidents. Integrating CTI and OSINT among the tools used by the incident response (IR) team enables a proactive response to security events. Awareness of the threat landscape can reduce uncertainty in security decision-making and IR processes. Threat intelligence can be incorporated into all IR phases, including preparation, acquisition of tools and resources, implementation of controls on assets (based on risk assessments), detection and

analysis, event confirmation and scoping, containment, eradication and recovery, and finally, producing an organizational report detailing the causes of the event, cost calculations, and steps to prevent similar incidents in the future [31].

Integration of threat intelligence into international cybersecurity standards: Since 2022, the International Organization for Standardization (ISO) has updated the ISO/IEC 27001 International Standard for Information Security Management, adding a new control: Control 5.7 on **threat intelligence**, in Annex A. Organizations are expected to demonstrate a process for collecting and analyzing threat intelligence to understand their threat environment. They must show evidence of collecting relevant threat-related information, analyzing it, and producing an up-to-date threat assessment (ISO Organization, 2022). Additionally, the National Institute of Standards and Technology (NIST) recommends incorporating cyber threat intelligence systems in the identification phase, and in the updated guidance released in February 2024, it also recommends integrating these systems into the detection phase [34], [62].

A supporting tool for cybersecurity and protection: OSINT and CTI can be used to analyze cyberattacks, manage cyber investigations, conduct digital forensics within the organization, and support organizational decision-making processes. Additionally, through data mining techniques, OSINT tools help analyze attacks, correlate between different incidents, and support effective defense and immediate response strategies [63].

Early threat detection: OSINT enables the identification of threat indicators in advance, allowing cybersecurity teams to prepare for future incidents, such as detecting data leaks, identifying spam or phishing campaigns, including spear phishing, and malware detection [64].

B. APPLICATIONS IN THE PUBLIC AND PRIVATE SECTORS

The public sector

The public sector is defined as the part of the economy that is controlled or owned by the government [65]. It provides services that contribute to social welfare, such as law enforcement, national defense, public transportation, infrastructure, education, and healthcare. Unlike the private sector, the public sector does not aim to profit from its services. OSINT applications specific to the public sector include military intelligence, the prevention of security threats and terrorism, political analysis, and academic research. In law enforcement, OSINT supports efforts against organized crime, the early identification of criminal intent, and even real-time detection of illegal activities.

Law enforcement agencies around the world use OSINT sources to protect citizens from abuse, sexual violence, identity theft, and other crimes. Monitoring of social media platforms and criminal networks is conducted based on keywords and images, which can help prevent crimes before they occur. For example, using OSINT techniques, authorities gather information on criminals (persons of interest) and compile complete profiles for each one [41]. This intelligence enables the collection of data on organized crime, early identification of criminal intent, and even real-time detection of illegal activities [66], [67].

Countering terrorism through OSINT: In recent years, terrorists have leveraged the digital space and social media to carry out malicious plans, using these platforms for propaganda, training, and advancing their objectives. Terrorism affects states across political, economic, religious, and social domains, becoming a growing global concern. The nature of a terrorist organization determines its attack types, target countries, target groups, and attack characteristics. OSINT offers an effective solution by extracting intelligence through data collection, enrichment, and knowledge inference [10]. Defense and security agencies must adapt to this reality by establishing OSINT units staffed with experts to gather intelligence on terrorist activities. OSINT is also recognized as a military reconnaissance and collection tool [9]. Informative sources like newspapers, news websites, and reports often include updates on terrorist events and sometimes real-time information about terrorist groups and actors. In some cases, terrorist organizations themselves post propaganda and incitement. Given the widespread reach of social media, terrorist events can also be identified by the spike in online activity and shifts in public sentiment [10].

Intelligence surveillance and reconnaissance (ISR): Every country collects information on other nations in the military and intelligence domains. The total volume of information gathered in this context is referred to as intelligence surveillance and reconnaissance (ISR). OSINT serves as a foundation for the methods used to collect and analyze this information [41], [68].

Using Open Sources to Corroborate Classified Leads: One of the strategic advantages of OSINT is its low classification level, which enables broad use and dissemination of intelligence outputs. In some cases, OSINT investigations can be designed to validate or reconstruct insights originally derived from classified sources, allowing them to be used in public or inter-agency contexts. For example, Bellingcat, an open-source investigative collective founded by former intelligence and research professionals, has conducted detailed investigations using publicly available data only. These investigations can support or mirror classified assessments without disclosing sensitive sources or methods. This capability allows intelligence agencies, security services, and international organizations to reference findings based on open sources when they are unable to directly share classified information [67], [68], [69].

The private sector

The private sector is defined as the part of the economy that is not controlled or owned by the government [70]. It includes entities from the financial sector such as banks, investment funds, real estate firms, rating agencies, and insurance companies. In developing countries, the private sector plays a significant economic role, providing nine out of ten jobs [71], and is responsible for 70–85% of the total global

TABLE 4. Comparison of OSINT applications in the public and private sectors, with areas of overlap.

Dimension	Public sector	Private sector	Shared / Collaborative Areas
Goals	National security, law enforcement, counterterrorism	Business continuity, cybersecurity, competitive analysis	Critical infrastructure protection, threat intelligence sharing, incident response
Resources	Government funding, access to classified information	Budget constraints, limited to open sources	Partnerships with vendors, public-private task forces
Access to Data	Classified and governmental databases	Commercial and open-source data	Joint threat intel platforms, ISACs/ISAOs (e.g., FS-ISAC), CERT collaboration
Challenges	Bureaucracy, regulatory compliance, ethics	Data validation, lack of trust, legal exposure	Data quality, cross-sector trust building, compliance with privacy laws (e.g., GDPR, HIPAA)
Organizational culture	Hierarchical, inter-agency collaboration	Agile, market-driven, innovation-oriented	Cross-sector information-sharing frameworks, multi-stakeholder cyber drills and simulations

investment [72]. OSINT applications unique to the private sector include business intelligence, analysis of public opinions and social attitudes, and analysis of user interests and preferences [67].

Use by business corporations: For corporations, information is powerful. Businesses use OSINT sources to explore new markets, monitor competitors' activities, plan marketing strategies, and anticipate factors that may impact current operations or future growth. In the past, OSINT was mainly used by large corporations with substantial budgets, however, today, due to the widespread availability of online information, even small businesses with limited resources can utilize OSINT and incorporate the resulting insights into their business strategies [41].

Table 4 summarizes the distinguishing characteristics of the applications of OSINT in the public and private sectors.

While Table 4 outlines key differences and areas of overlap in OSINT use across the public and private sectors, it is important to recognize that in practice, these domains often cooperate closely, especially in areas such as national security, critical infrastructure protection, and cybersecurity. Many of these distinctions are contextual rather than absolute, and joint frameworks have emerged to support information sharing and collaborative analysis. An example of a well-developed and structured OSINT ecosystem can be seen in the United States. The following section describes how OSINT is implemented across government, intelligence, and private sectors in the U.S., highlighting how policy, operational processes, and national strategy support integrated open-source intelligence capabilities.

C. CASE STUDY: OSINT IMPLEMENTATION IN THE UNITED STATES

The United States is considered a global leader in OSINT because of technological, institutional, and operational advantages. It maintains advanced capabilities in data analysis, tool development, and automation, driven by its leadership in information technology. A well-resourced intelligence community supports these efforts with significant infrastructure and expertise dedicated to open-source collection and analysis. Substantial investment in OSINT-related research and development further enhances U.S. capacity in this field. Additionally, the abundance of publicly available information from social media, public records, news media, and academic sources provides a rich foundation for open-source intelligence efforts. The U.S. also benefits from international partnerships and a dynamic private sector that drives innovation in software, data analytics, and cybersecurity. Education and training programs have been established to build a skilled OSINT workforce and improve operational capabilities [7], [39], [68], [73], [74].

1) LEGISLATIVE FRAMEWORK–THE PATRIOT ACT

After the September 11, 2001 attacks, U.S. intelligence and law enforcement recognized the need for improved information sharing. The Patriot Act and Homeland Security Act, enacted shortly afterward, removed legal barriers that had previously hindered cooperation between intelligence and law enforcement agencies. These laws aim to enhance national security by improving coordination, enabling the exchange of intelligence, and consolidating efforts under the newly created Department of Homeland Security [75], [76].

2) STRATEGIC DEVELOPMENTS–THE 9/11 COMMISSION AND DoD OSINT STRATEGY

The 9/11 Commission's 2004 report identified significant gaps in the U.S. government's ability to fully utilize intelligence from all sources, including OSINT. In response, the Secretary of Defense initiated the development of a formal

defense open-source intelligence strategy to integrate OSINT more effectively into national security operations [6], [77].

3) ORGANIZATIONAL IMPLEMENTATION–INTELLIGENCE COMMUNITY AND KEY AGENCIES

The U.S. Intelligence Community (IC) integrates OSINT across several key agencies. CIA's Open-Source Enterprise (OSE) focuses on collecting and analyzing publicly available information for national security purposes. The NSA incorporates OSINT into signals intelligence workflows, while the Defense Intelligence Agency (DIA) leverages OSINT alongside other sources to support military operations and strategic planning. These agencies use OSINT to enhance situational awareness, monitor threats, and support decision-making at all levels [7], [39], [68], [74], [77].

4) PUBLIC-PRIVATE COLLABORATION–OPERATIONAL INTEGRATION

Public-private cooperation is a central feature of the U.S. OSINT model. Agencies such as the Department of Homeland Security (DHS) and the Cybersecurity and Infrastructure Security Agency (CISA) partner with private sector companies to share threat intelligence and technological tools. Programs such as Automated Indicator Sharing (AIS) enable real-time exchange of cyber threat indicators between government and industry, improving the nation's ability to detect and respond to emerging threats. Technology firms contribute advanced tools and analytics, whereas public agencies offer coordination, regulatory guidance, and access to broader intelligence networks [77].

Despite these advances, challenges persist. Ensuring data reliability, managing legal risks, and maintaining operational security require constant attention. Nonetheless, the U.S. model demonstrates that strategic investment, cross-sector collaboration, and policy support can create a scalable and effective OSINT ecosystem.

D. ADDITIONAL APPLICATIONS

1) USE BY INDIVIDUALS

In recent years, anyone searching for information using an internet search engine is effectively using OSINT technology to find publicly available online data. Private users "collect" OSINT data for everyday purposes, such as for studies, work, research, or personal interests [18].

2) DEEPFAKE

In recent years, there has been an increase in deepfakes, fake news, deepfake technology, and artificial intelligence (AI), all of which have introduced powerful new methods for spreading disinformation. Deepfakes involve synthetically altered images, videos, audio clips, or text to make them appear as someone or something else. Advances in AI have made deepfake technology cheaper and more accessible, enabling users without technical backgrounds to create realistic fake videos within minutes. This raises concern about the growing threat posed by both foreign and domestic disinformation campaigns. Such false content is distributed online to influence public opinion on politics, news, protests, vaccine effectiveness (e.g., COVID-19), and even national security. OSINT tools can help map networks to identify creators and distributors of deepfake content, aiding content verification., For example, the BBC trains journalists to use OSINT methods to validate the authenticity of published content. There is increasing recognition that deepfake-driven disinformation campaigns pose significant risks [68], [78], [79].

3) MISUSE OF OSINT DATA BY TERRORISTS AND HACKERS

An attacker can select a target based on an online profile, conduct extensive intelligence gathering, and then conduct cybercrimes such as hacking into the target's computer, stealing or spreading confidential or false information, sending malware or spam, phishing, and launching DoS attacks. Most cybercrimes aim to steal sensitive or financial data, extort money, or commit identity theft. More severe cybercrimes, particularly those affecting national security, include disrupting critical network operations, overloading systems, exploiting vulnerabilities to disable services, and spreading false information to provoke political instability [41].

4) EXTRACTION OF IMPLICIT (NON-EXPLICIT) INFORMATION

OSINT tools enable the extraction of implicit or "unexplained" information from raw data using advanced analytical methods, including correlation analysis, metadata examination, geo-temporal mapping, and machine learning. These tools help cybersecurity professionals infer indirect evidence, detect anomalies, and identify hidden threat vectors, thereby contributing to a proactive and contextual threat intelligence framework. Additionally, such information has significant value and application in various civil domains, such as sentiment and opinion analysis, interaction tracking, message analysis, and identifying users' interests and preferences [80].

E. PRACTICAL STRATEGIES FOR PROACTIVE CYBERSECURITY USING OSINT

Following the review of OSINT applications in different sectors and environments, this section outlines practical strategies that organizations can adopt to implement OSINT to support proactive cybersecurity. These recommendations translate the insights discussed in the previous sections into concrete, operational practices:

- **Align with ISO/IEC 27001:2022 Requirements:** Beginning in 2025, organizations must implement cyber threat intelligence measures to comply with updated ISO/IEC 27001 standards. OSINT can fulfill these requirements by supporting early threat detection, external risk identification, and intelligence-based risk management.
- **Establish a Defined OSINT Workflow:** Organizations should design a structured OSINT process, from

collection planning through analysis to dissemination, to ensure consistency and strategic alignment. This allows OSINT to be systematically integrated with security operations.

- **Integrate OSINT into Existing Security Infrastructure:** OSINT tools and threat feeds can be incorporated into platforms such as Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR), thereby improving the context and speed of incident response.
- **Select Purpose-Fit OSINT Tools:** Organizations should deploy targeted tools depending on their use case. Examples include, Shodan or Censys for exposed infrastructure, Maltego for link analysis, and SpiderFoot or Recon-ng for broad-spectrum reconnaissance.
- **Assign OSINT Roles and Responsibilities:** Defining whether OSINT tasks are handled by internal analysts, managed security service providers (MSSPs), or dedicated threat intelligence teams is essential for operational clarity and effectiveness.
- **Validate and Escalate Findings with Clear Procedures:** Given the open nature of OSINT, verification is critical. Organizations should implement workflows for data validation, triage, and escalation to avoid acting on incomplete or inaccurate information.
- **Comply with Legal and Ethical Standards:** OSINT use must adhere to relevant data protection regulations and avoid unethical practices. This includes handling personal data responsibly and avoiding deceptive investigative methods unless explicitly authorized.

By applying these strategies, organizations can transition from reactive to proactive cybersecurity postures, using OSINT not only to monitor threats but also to anticipate and mitigate them.

V. OSINT: ADVANTAGES AND CHALLENGES

While OSINT supports proactive cybersecurity by providing early threat visibility and low-cost intelligence, its effectiveness also depends on understanding its limitations. This chapter outlines the advantages that enable preventive action, as well as the operational and ethical challenges that must be addressed to make OSINT a reliable part of proactive defense strategies.

A. ADVANTAGES

- **Rapid/real-time data collection:** Information from open sources is received quickly and almost in real time. Searching a wide range of OSINT sources increases the likelihood of obtaining the desired data and ensures rapid access to the required information [41].
- **Massive volume of available information**: In the past, OSINT experts aimed to locate hidden or hard-to-find information on the internet. Today, the volume of data is growing at an unprecedented rate due to increased internet use. Therefore, the challenge for OSINT systems is to identify relevant information that can be translated into explicit and useful knowledge [81].
- **Secure collection of large amounts of data:** Data collected from open sources is accessible to everyone, the collection is legal, and the security risk is low. The data is received securely, and the reliability of the information can be increased by processing a significant amount of data [41].
- **Multilingual content:** Hackers worldwide communicate in various languages, such as English, Chinese, Russian, Persian, Hindi, and Arabic. Decoding and analyzing multiple languages present a challenge during the data collection and enrichment phases. Many languages with national status have available transcription APIs. However, some attackers use rare languages spoken in remote regions, that are very difficult to decode. It is necessary to identify these regional languages and build translation corpora with the assistance of language experts [10].
- **High computing capability:** The growth in the use of OSINT systems is attributed to advancements in computer architecture, processors, and graphics processing units (GPUs). These developments enable the execution of labor-intensive tasks in terms of data collection, processing, analysis, and storage [82]. Owing to its improved computing power, OSINT can now be applied using advanced processing and analysis techniques to large volumes of public information, integrating numerous datasets and identifying various types of connections and patterns.
- **Big data and machine learning:** The rapid development of data analysis and mining techniques, as well as machine learning algorithms, can make investigation and decision-making processes more automated and efficient [82]. These techniques allow the identification of complex correlations that are not naturally predictable by humans. This aspect will be a key factor in future OSINT activities, marking the difference between human-driven research and AI-based research. The integration of these techniques will improve the collection and analysis process and support more accurate investigations that align with defined objectives. Additionally, government counterintelligence agencies can leverage these techniques to enhance information quality, thereby strengthening the fight against terrorist organizations [83].
- **Complementary data types:** OSINT systems can be supplemented with additional types of information [84]. The system architecture is sufficiently open to accept data that are not directly obtained from open sources. This means that the OSINT system can be even more effective when external pieces of information are added to support investigations. For example, law enforcement agencies can benefit from citizen cooperation to contribute to OSINT searches. Cyber intelligence services can leverage classified information on cybercriminals or

incidents to enrich OSINT investigations. Even private individuals can combine OSINT with social engineering techniques to create a profile on their target.

- **Flexible purpose and broad scope:** Owing to the comprehensive nature of OSINT systems, investigations can be expanded to many different purposes while collecting pieces of information from across the internet. These systems can be used in research across fields such as economics, psychology, strategy, journalism, employment, and security. In particular, the benefits in the areas of crime and cybersecurity should be emphasized, such as monitoring suspicious individuals or dangerous groups, identifying influential profiles associated with radicalization, investigating social trends, supporting the attribution of cyberattacks and crimes, and improving digital forensic diagnostics [66], [85].
- **Low-cost data collection:** Compared with the high costs of training agents in HUMINT or the costs of collecting data using satellites and uncrewed aerial vehicles (UAVs) in TECHINT, OSINT offers a significant advantage in obtaining data at a low cost [41].

B. CHALLENGES

As noted, OSINT systems face numerous challenges. Below is a partial list of these challenges:

- **Excessive volume of information:** The more data that is collected, the harder it becomes to extract reliable information. OSINT tools gather data from numerous sources; therefore, time and effort are required to identify false information and extract reliable data [41].
- **Complexity of data management:** The volume of data is vast, and as a result, the challenge of handling it efficiently and effectively is significant [86]. Although there are benefits to broad data collection, OSINT systems must consider as much information as possible. In addition, systems must employ advanced techniques and substantial resources to ensure high-quality collection, processing, and analysis.
- **Unstructured data:** Publicly available internet information is not organized. This means that the data collected by OSINT is highly heterogeneous, making classification difficult. Therefore, it is necessary to integrate and analyze these data to generate relevant connections and knowledge [87]. In this sense, OSINT requires mechanisms such as data mining, natural language processing (NLP), and text analysis to organize unstructured information and extract its value.
- **Fabricated information:** Social networks and media channels are flooded with subjective opinions, fake news, and fabricated stories [87]. Therefore, the existence of inaccurate information must be considered when implementing OSINT mechanisms, and the information collected should be verified and checked as much as possible. OSINT activities should always rely on credible information and follow trustworthy investigative leads in order to ensure positive and convincing outcomes [88], [89].
- **A cornerstone of cybercrime:** Attackers with malicious intent can, like anyone else, access data from open sources and misuse it. Therefore, research is needed on security requirements and technological measures that can reduce the damage caused by such crimes [41]. An attacker may collect data regarding a target, their family, and their home in order to strike at the most opportune time. A blackmailer may publish private and personal information about the victim if the ransom is not paid. Developers must take these aspects into consideration when implementing OSINT tools. It is crucial that the most powerful OSINT tools remain accessible only to law enforcement agencies and intelligence services [42], [44].
- **Reliability of data sources:** The reliability and validity of information are key to successful OSINT investigations [90]. Ideally, the data collected should come from certified, verified, and reliable sources, such as official documents, scientific reports, and trustworthy media outlets [86]. In practice, however, OSINT can also collect data from unreliable or subjective sources, including social media content or manipulated media [91]. There are cases where misinformation obtained from unreliable sources can be used. for example, when OSINT researchers have no "clue" in the investigation, this information can help in extracting initial information about individuals, groups, or companies [92].
- **Strong ethical/legal considerations:** Significant concerns about privacy, dignity, and personal integrity arise with the development of OSINT systems [3], [93]. The question of whether OSINT constitutes an ethical issue typically falls within the domain of intelligence collection ethics [94]. Although open-source information is publicly accessible, OSINT can reveal information that is not explicitly published online. The disclosed results should respect users' privacy and avoid exposing intimate or personal issues [95]. Relevant current regulations should be considered, such as the General Data Protection Regulation (GDPR), which is the European Union's framework for data privacy protection [96]. Through such data, it is possible to infer aspects such as sexual orientation, religious beliefs, political affiliation, or suspicious behavior. This exposure process can be problematic in many countries; therefore, the scope of OSINT-based searches should be limited to the search target, and the collected data must be secured at a high level [8].
- **Organizational perception and biases within intelligence agencies:** In the organizational culture of intelligence agencies, the value of data collected via OSINT is often underestimated. Such data is not "valued" since anyone can access and use it. Intelligence agencies typically operate using internal (classified) computer networks because of security concerns,

TABLE 5. Summary of key references.

Author(s)	Year	Focus area	Key findings / Contribution	Relevance to this study
Hwang et al.	2022	OSINT capabilities and limitations	Outlined OSINT's strengths (real-time data, low-cost collection) and risks (data overload)	Supports benefits and limitations of OSINT systems
Nicart et al.	2017	Data volume and relevance in OSINT	Addressed challenges of extracting useful intelligence from massive open data volumes	Frames the filtering and processing challenge in OSINT
Bello-Orgaz et al.	2016	Unstructured data in social media	Discussed OSINT's struggle with unstructured, heterogeneous data	Informs the data analysis constraints faced in OSINT
Gandomi & Haider	2015	Big data analytics and machine learning	Described the role of ML in deriving value from large datasets	Supports use of ML in OSINT automation
Larsen et al.	2017	Security and criminology use cases	Showed OSINT's role in digital forensics and social risk profiling	Demonstrates cross-sector utility of OSINT
Tabatabaei & Wells	2016	Strategic intelligence applications	Highlighted OSINT's role in attribution and geopolitical analysis	Strengthens theoretical argument for OSINT in strategic defense
Gong et al.	2018	Source reliability in OSINT	Emphasized importance of source validation in open data investigations	Supports quality and trust concerns in OSINT usage
Weinbaum et al.	2022	Government and civilian OSINT practices	Analyzed OSINT integration in public and private sectors in the U.S.	Supports cross-sector application and implementation strategy
Pastor-Galindo et al.	2020	Multimedia and NLP in OSINT	Reviewed NLP and data mining for intelligence from unstructured data	Supports automation and semantic analysis in OSINT pipelines
Chaudhary & Bansal	2022	Multilingual processing in OSINT	Identified linguistic challenges in OSINT data processing	Supports discussion of NLP and language barriers
Taddeo & Floridi	2018	Ethical and governance frameworks	Discussed ethical challenges in cyber operations and OSINT	Replaces older gray references and supports ethical dimensions of OSINT
Bradshaw et al.	2022	Legal and regulatory challenges	Analyzed cyber law frameworks applicable to OSINT, including GDPR	Replaces Simola 2020 and supports legal limitations in EU context

which restrict the use and collection of data via the internet. Consequently, analysts in intelligence agencies tend to show a passive attitude toward the use of OSINT data. Information security experts are working to design methods for the safe and unrestricted use of such information while addressing related security issues [41].

VI. METHODOLOGY

This study adopts a structured literature review methodology, drawing on elements from the Systematic Literature Review (SLR) approach. The goal of this method is to provide a transparent and comprehensive overview of the current state of knowledge regarding Open-Source Intelligence (OSINT) in cybersecurity. The review process was designed to support the central argument, that OSINT plays a critical role in enabling proactive cybersecurity through early threat detection, strategic planning, and informed decision-making. This core concept guided the identification, selection, and analysis of the literature.

Academic sources were identified through systematic keyword searches in established databases including the Web of Science, Scopus, IEEE Xplore, and Google Scholar. The search strategy focused on combinations of keywords such as "OSINT AND cybersecurity," "open-source intelligence AND threat detection," "cyber threat intelligence," "OSINT AND automation," "OSINT AND proactive defense," and "NLP AND OSINT." To ensure the material reflects the evolving nature of cybersecurity, the search prioritized publications from the last ten years, with special attention given to articles published in 2024 and 2025. A summary of the key references identified in this process is presented in Table 5.

The inclusion criteria focused on peer-reviewed academic publications that address OSINT's applications, limitations, technologies, legal and ethical considerations, and contributions to cyber resilience. Studies were included if they offered empirical data, presented technical frameworks, analyzed case studies, or proposed methodologies relevant to OSINT's use in cyber threat detection, intelligence production, or organizational defense. Articles unrelated to digital

security or focused exclusively on military operations without cyber relevance were excluded.

To provide a practical context for this evolving and operationally oriented domain, select gray literature was also incorporated. These sources included industry reports, government publications, military manuals, and cybersecurity standards. In particular, documents such as the ISO/IEC 27001 standard were used to highlight formal requirements for cyber threat intelligence implementation. The inclusion of gray literature was limited to sources that are authoritative and relevant to current OSINT practices. Wherever possible, these materials were supplemented or corroborated by peer-reviewed academic studies to ensure a balanced and academically rigorous foundation.

Following source selection, the material was reviewed and categorized into five key thematic areas: (1) OSINT conceptual foundations and definitions; (2) technical tools and methods, including Natural Language Processing (NLP) and automation; (3) cybersecurity applications across the public and private sectors; (4) operational, ethical, and legal challenges; and (5) proactive defense strategies and future research directions. This thematic classification enabled an organized synthesis of diverse materials and helped connect individual findings to the overarching research argument.

Throughout the manuscript, evidence from the literature was integrated to support the structure and argumentation. Emphasis was placed on identifying both recurring themes and gaps in the literature, particularly in the context of OSINT's role in enhancing proactive cybersecurity. The revised manuscript also includes recent studies that reflect the state of the field over the past year and improves the balance between academic sources and essential operational references. This structured approach ensures that the review is comprehensive, focused, and aligned with the theoretical and practical contributions of this study.

VII. DISCUSSION

The discussion brings together the key findings of the study and places them within the broader context of proactive cybersecurity. It explores how technological advances, analytic methods, and ethical considerations influence OSINT's effectiveness as a forward-looking security tool. New insights are also presented on how OSINT integrates with modern threat detection systems.

The literature reviewed in this study highlights the evolving role of OSINT as a critical component in modern cybersecurity strategies. OSINT is not merely a supplementary tool but is increasingly used as a proactive mechanism for threat detection, situational awareness, and decision-making. Its integration within cyber threat intelligence (CTI) processes underscores its ability to provide timely, relevant, and contextual information derived from publicly available sources. The versatility of OSINT applications across sectors is a central theme emerging from the literature. Governmental, military, and private organizations all utilize OSINT for diverse purposes, ranging from cybercrime prevention and national security to fraud detection and vulnerability assessments. This broad applicability reflects OSINT's adaptability to different operational environments and threat models. However, several challenges limit its potential. The reliability and verification of data collected from open sources remain significant concerns, particularly in relation to misinformation, data manipulation, and source credibility. Legal and ethical issues further complicate its use, particularly when OSINT intersects with privacy rights and regulatory frameworks. Moreover, the vast volume of unstructured data imposes analytical burdens that require both human expertise and technological automation.

While the traditional six-phases intelligence cycle (direction, collection, processing, analysis, dissemination, and feedback) is a foundational model for intelligence work, it has been criticized for being too linear and inflexible for fast-paced cyber environments. Cyber threat intelligence (CTI) often requires rapid collection, real-time analysis, and immediate response, which are not easily supported by a sequential process. For example, in many cyber operations, dissemination may happen before full analysis is complete, or initial direction may be redefined mid-process as new threats emerge. To address these challenges, more adaptive models have been proposed. One such model is the **continuous feedback loop**, where each phase of the intelligence cycle informs and updates the others dynamically. Another model was proposed by **Rand Corporation** and has four stages: collection, processing, exploitation, and production, which is adapted for artificial intelligence-based OSINT systems [2], [97]. Another approach is the **Cyber Kill Chain** (Lockheed Martin), which focuses on the stages of a cyberattack and emphasizes proactive detection and disruption of adversary activities. Additionally, frameworks such as **F3EAD** (Find, Fix, Finish, Exploit, Analyze, Disseminate), originally developed in military contexts, offer a more action-oriented structure that is often used in cyber threat hunting [98]. These modern approaches are better suited to the iterative and time-sensitive nature of cyber intelligence work. By integrating these models, CTI teams can increase their responsiveness and adapt more effectively to changing threat landscapes.

Expanded Ethical Considerations in OSINT Use: While privacy is a well-recognized ethical concern in the use of OSINT, several emerging challenges also require attention. One such issue is the **weaponization of OSINT**, where publicly available data is used for **targeted harassment**, doxxing, or intimidation campaigns. Additionally, **data voids** and gaps in online information that can be intentionally filled with misleading or harmful content, pose a risk for disinformation operations. Another ethical gray area involves the use of **"sock puppet" accounts**, where investigators create false online identities to gain access to restricted information or communities. While this technique can enhance investigations, it also raises questions about deception, consent, and compliance with platform policies. These concerns highlight the importance of developing ethical guidelines and oversight

mechanisms for the responsible use of OSINT in both public and private sectors.

Integration of Machine Learning and Big Data Architectures in OSINT: The application of OSINT in cybersecurity increasingly relies on big data infrastructures and machine learning algorithms to process large volumes of unstructured and multi-source data. Common architectures include distributed processing frameworks, such as Apache Hadoop and Apache Spark, which enable real-time analysis and scalable data handling. Machine learning models are used at various stages, and classification algorithms such as Support Vector Machines (SVM) and Random Forests help in identifying threat categories. Clustering techniques such as k-means are used to detect anomalies in behavioral data, and deep learning models such as LSTM and BERT are applied in natural language processing tasks, including sentiment analysis and topic modeling. These tools enable the automated identification of threats across dynamic OSINT sources, such as social media, blogs, and forums. However, OSINT data is often noisy, incomplete, or intentionally deceptive, making preprocessing steps, such as deduplication, normalization, and validation, critical. Data fusion methods, whether through early integration of raw inputs or late integration of model outputs, are essential to synthesizing a coherent intelligence output from diverse sources. These technical approaches improve the speed, accuracy, and operational relevance of intelligence derived from open sources.

This study makes a dual contribution to the field of cybersecurity. **Theoretically,** it positions OSINT as a core component of CTI frameworks, presenting it as a dynamic, proactive mechanism that supports early threat detection and contributes to organizational cyber resilience. It bridges multiple fields, including intelligence research, cyber risk management, and information systems, into a unified understanding of OSINT's evolving role. It also emphasizes OSINT's function in both public and private sectors, offering a foundation for future research. **Practically,** the study provides a structured and actionable perspective on how organizations can leverage OSINT to enhance their cybersecurity posture. By synthesizing real-world applications from government, military, and private sectors, it outlines operational strategies for integrating OSINT in threat detection, fraud investigation, dark web monitoring, compliance with cyber standards, and organizational vulnerability assessment. The study also discusses recent advances in OSINT automation, multimedia analysis, and crowdsourced intelligence, offering decision-makers a toolkit of proactive defense methods and technologies. Security professionals and policymakers can use these insights to strengthen incident response protocols, improve information-sharing networks, and guide investment in OSINT platforms. **The importance of this study** lies in its focus on OSINT as a critical yet underutilized resource in modern cybersecurity. As cyber threats grow more complex, fast-moving, and globally distributed, the ability to extract real-time, relevant intelligence from publicly available sources becomes essential. This study addresses a clear research gap by providing a consolidated review of OSINT's role in enhancing cyber resilience and offers both strategic and technical perspectives. It encourages the cybersecurity community to reconsider how intelligence is collected, validated, and operationalized in an era of information overload and increasing digital complexity. This reinforces the article's conceptual contribution by positioning OSINT as a key driver of proactive cybersecurity strategy in both research and practice.

VIII. CONCLUSION

The concluding chapter reaffirms the value of OSINT as a foundation for proactive cybersecurity. It summarizes how the tools, methods, and practices discussed throughout the study contribute to early threat identification and response. The chapter also outlines future research directions to enhance OSINT's role in shaping adaptive and preventive cyber defense.

This study presents a comprehensive conceptual review of OSINT as a foundational component of proactive cybersecurity. The sources confirm that OSINT plays a strategic role in enhancing cyber threat intelligence, facilitating early threat detection, and supporting organizational resilience in dynamic cyber environments. By collecting data from publicly accessible sources, OSINT enables fast and effective insights into emerging threats, attacker behavior, and system vulnerabilities. Its integration into cybersecurity frameworks empowers both public and private entities to anticipate and mitigate risks before they materialize. However, the practical application of OSINT is constrained by challenges related to data quality, legal compliance, and ethical considerations. In conclusion, OSINT is a significant tool for advancing proactive cybersecurity measures. Nevertheless, to fully realize its potential, further efforts are required to establish operational standards, ensure ethical compliance, and enhance analytical capabilities through automation and training.

A. FUTURE RESEARCH DIRECTIONS

This study presents the strategic relevance of OSINT in cybersecurity. Several areas warrant further academic investigation to advance both theory and practice:

Enhancing cyber resilience: There is growing interest in understanding how OSINT can support cyber resilience at both the organizational and national levels. Longitudinal studies assessing OSINT's contribution to building resilience, responding to crises, and ensuring regulatory compliance could offer valuable insights to the field. Research question: *How does the integration of OSINT into organizational cybersecurity strategies improve resilience against different types of cyberattacks over time?*

Dark web monitoring: The dark web is a significant source of cybercrime and illegal activity. Innovations in this area include the development of advanced techniques for scanning, monitoring, analyzing, and indexing the dark web. These advancements will enable researchers and analysts to

collect and analyze data from hidden services, illicit marketplaces, forums, and other dark web platforms. Research question: *How effective is NLP topic modeling in identifying emerging threats in non-English forums and marketplaces on the dark web?*

Multimedia data collection and analysis: As the use of multimedia content (images, videos, audio) increases across online platforms, new methods are required for data collection, analysis, and intelligence extraction from these sources. Examples include improvements in image recognition, video analysis, audio processing, speech-to-text conversion, and metadata extraction techniques. Innovations in multimedia data collection enhance the ability to detect visual threats, identify individuals or objects, and derive meaningful insights from multimedia content. Research question: *What is the accuracy and operational value of automated multimedia analysis techniques (e.g., image recognition, speech-to-text) in extracting actionable intelligence from OSINT sources?*

Real-time data collection: Real-time data collection is becoming increasingly important. Innovations involve the development of tools and methodologies for collecting and analyzing real-time data, enabling faster identification and response to threats, incidents, or emerging events. Real-time data collection techniques may include monitoring social media updates, online sources, and other new platforms, with the goal of quickly identifying and analyzing relevant information. Research question: *What are the technical and operational challenges in implementing real-time OSINT collection systems for early threat detection in fast-changing online environments?*

Collaboration and crowdsourcing: Innovations in open-source data collection include collaborative and crowdsourced approaches. Enhancing and leveraging the power of collective intelligence for collecting and analyzing data from open sources. Crowdsourcing platforms can be used to engage a community of volunteers and experts to contribute to data collection efforts, verify information, and provide insights and analysis. Collaboration and crowdsourcing improve the scalability, diversity, and accuracy of data collection [74]. Research question: *How can structured crowdsourcing platforms be used to validate and enrich OSINT data without compromising data reliability or security?*

The Transition to Web 3.0: The foundation of the internet was Web 1.0, which included text and static images. The next generation, Web 2.0, was defined by interactivity and social media. The third generation of the internet, Web 3.0 also known as Web3 is designed to be decentralized, open to all (with a bottom-up design), built on blockchain technologies, and includes features and developments related to the semantic web. This network is structured as decentralized networks of numerous nodes (servers), forming a crypto-economic protocol. The advantages of this network include verifiability, reliability, decentralization, the absence of a central authority, robustness, and built-in payment systems. In the next generation of the internet, machine learning and natural language processing technologies will be dominant, and improving the efficiency of these methods for OSINT purposes is necessary [2], [99]. Therefore, it is essential to understand the opportunities within this network and to adapt OSINT system readiness accordingly. Research question: *What role can decentralize Web 3.0 technologies play in enhancing the effectiveness and reliability of OSINT systems, particularly in hostile or censored information environments?*

Predicting the next major event: According to the Black Swan theory developed by philosopher Nassim Taleb, a black swan is an extraordinary event with a significant impact that appears unforeseeable in advance, yet historically, such events should be expected occasionally [100]. Effective OSINT systems will attempt to forecast such events by monitoring and analyzing sources and social networks, identifying trends and changes, and applying analytics tools to big data, data processing, and content analysis. Existing tools can assist in tracking internet trends and developments, such as Google Trends. Research question: *Can OSINT-based trend analysis combined with predictive analytics improve early warning capabilities for unexpected, high-impact cyber or geopolitical events?*

Legal aspects: The legal and ethical dimensions of OSINT have been scarcely studied in academic literature. Future research should address regulatory compliance, privacy implications, and the ethical boundaries of using data collected from open sources, particularly when it involves personal or sensitive information. Research question: *What legal and ethical frameworks are required to ensure responsible use of OSINT that involves personal or sensitive data, while maintaining operational utility in cybersecurity operations?*

REFERENCES

[1] Check Point Research. *Cyber Security Report 2025 | Check Point Software*. Accessed: Apr. 15, 2025. [Online]. Available: https://www.checkpoint.com/security-report/

[2] H. J. Williams and I. Blum, "Defining second generation open source intelligence (OSINT) for the defense enterprise," RAND Corporation, May 2018, doi: 10.7249/RR1964.

[3] J.-J. Oerlemans and S. Langenhuijzen, "Balancing national security and privacy: Examining the use of commercially available information in OSINT practices," *Int. J. Intell. Counter Intell.*, vol. 38, no. 2, pp. 579–597, Apr. 2025, doi: 10.1080/08850607.2024.2387850.

[4] G.-T. Ungureanu, "Open source intelligence (OSINT). The way ahead," *J. Defense Resour. Manage.*, vol. 12, no. 1, pp. 177–200, 2021. [Online]. Available: https://www.ceeol.com/search/article-detail?id=960765

[5] L. Block, "The long history of OSINT," *J. Intell. Hist.*, vol. 23, no. 2, pp. 95–109, May 2024, doi: 10.1080/16161262.2023.2224091.

[6] U.S. Department of Defense. *U.S.C. Title 50-War and National Defense*. Accessed: Jun. 17, 2023. [Online]. Available: https://www.govinfo.gov/content/pkg/USCODE-2011-title50/html/USCODE-2011-title50-chap15-subchapI-sec403-5.htm#:%7E:text=%E2%80%9C(1)%20Open%2Dsource,addressing%20a%20specific%20intelligence%20requirement

[7] U.S. Army. (2012). *Open-Source Intelligence ATP 2-22.9*. Accessed: Jul. 12, 2023. [Online]. Available: https://irp.fas.org/doddir/army/atp2-22-9.pdf

[8] M. Zwanenburg, "The use of OSINT for military operations abroad under international humanitarian law and international human rights law," *Chin. J. Int. Law*, vol. 23, no. 3, pp. 403–442, Sep. 2024, doi: 10.1093/chinesejil/jmae027.

[9] A. Ziółkowska, "Open source intelligence (OSINT) as an element of military recon," *Secur. Defence Quart.*, vol. 19, no. 2, pp. 65–77, Jun. 2018, doi: 10.5604/01.3001.0012.1474.
[10] M. Chaudhary and D. Bansal, "Open source intelligence extraction for terrorism-related information: A review," *WIREs Data Mining Knowl. Discovery*, vol. 12, no. 5, p. 30, Sep. 2022, doi: 10.1002/widm.1473.
[11] J. R. G. Evangelista, R. J. Sassi, M. Romero, and D. Napolitano, "Systematic literature review to investigate the application of open source intelligence (OSINT) with artificial intelligence," *J. Appl. Secur. Res.*, vol. 16, no. 3, pp. 345–369, Jul. 2021, doi: 10.1080/19361610.2020.1761737.
[12] H4D. *Hacking for Defense*. Accessed: Jul. 9, 2023. [Online]. Available: https://www.h4d.us/
[13] B. Katz. (2020). *The Collection Edge: Harnessing Emerging Technologies for Intelligence Collection*. CSIS BRIEFS. Accessed: Jul. 9, 2023. [Online]. Available: https://www.csis.org/analysis/collection-edge-harnessing-emerging-technologies-intelligence-collection
[14] Statista. *Internet and Social Media Users in the World 2025 | Statista*. Accessed: Apr. 14, 2025. [Online]. Available: https://www.statista.com/statistics/617136/digital-population-worldwide/
[15] Statista. *Online Usage Per Minute 2024 | Statista*. Accessed: Apr. 14, 2025. [Online]. Available: https://www.statista.com/statistics/195140/new-user-generated-content-uploaded-by-users-per-minute/
[16] M. Nouh, J. R. C. Nurse, H. Webb, and M. Goldsmith, "Cybercrime investigators are users too! Understanding the socio-technical challenges faced by law enforcement," Dept. Comput. Sci., Univ. Oxford, Cornell Univ., New York, NY, USA, Tech. Rep., Feb. 2019. [Online]. Available: https://dx.doi.org/10.14722/usec.2019.23032
[17] P. Maciolek and G. Dobrowolski, "CLUO: Web-scale text mining system for open source intelligence purposes," *Comput. Sci.*, vol. 14, no. 1, pp. 45–62, 2013, doi: 10.7494/csci.2013.14.1.45.
[18] Statista. *Global Search Engine Market Share 2025 | Statista*. Accessed: Apr. 14, 2025. [Online]. Available: https://www.statista.com/statistics/1381664/worldwide-all-devices-market-share-of-search-engines/
[19] D. Chismon and M. Ruks. (2015). *Threat Intelligence: Collecting, Analysing, Evaluating*. Accessed: Jun. 27, 2023. [Online]. Available: https://nsarchive.gwu.edu/document/17212-united-kingdom-government-threat-intelligence
[20] R. M. Blank and P. D. Gallagher, "NIST-guide for conducting risk assessments," NIST Special Publication, Gaithersburg, MD, USA, Tech. Rep. NIST SP 800-30 Rev. 1, 2012, doi: 10.6028/NIST.SP.800-30r1.
[21] C. Johnson, L. Badger, D. Waltermire, J. Snyder, and C. Skorupka, "NIST special publication 800–150 guide to cyber threat information sharing," NIST Special Publication, Gaithersburg, MD, USA, Tech. Rep., 2016, doi: 10.6028/NIST.SP.800-150.
[22] Gartner Research. *Definition: Threat Intelligence*. Accessed: May 6, 2023. [Online]. Available: https://www.gartner.com/en/documents/2487216
[23] S. Barnum, "Standardizing cyber threat intelligence information with the structured threat information eXpression (STIX)," Mitre Corp., Bedford, MA, USA, Tech. Rep., May 2014. [Online]. Available: http://stixproject.github.io/about/STIX_Whitepaper_v1.1.pdf
[24] W. Tounsi and H. Rais, "A survey on technical threat intelligence in the age of sophisticated cyber attacks," *Comput. Secur.*, vol. 72, pp. 212–233, Jan. 2018, doi: 10.1016/j.cose.2017.09.001.
[25] D. Omand, "Social Media Intelligence (SOCMINT)," in *The Palgrave Handbook of Security, Risk and Intelligence*, Jul. 2017, pp. 355–371, doi: 10.1057/978-1-137-53675-4_20.
[26] I. Deliu, C. Leichter, and K. Franke, "Extracting cyber threat intelligence from hacker forums: Support vector machines versus convolutional neural networks," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2017, pp. 3648–3656, doi: 10.1109/BIGDATA.2017.8258359.
[27] D. Guido, "A case study of intelligence-driven defense," *IEEE Secur. Privacy*, vol. 9, no. 6, pp. 67–70, Nov. 2011, doi: 10.1109/MSP.2011.158.
[28] S. E. Dog, A. Tweed, L. Rouse, B. Chu, D. Qi, Y. Hu, J. Yang, and E. Al-Shaer, "Strategic cyber threat intelligence sharing: A case study of IDS logs," in *Proc. 25th Int. Conf. Comput. Commun. Netw. (ICCCN)*, Aug. 2016, pp. 1–6, doi: 10.1109/ICCCN.2016.7568578.
[29] V. G. Li, M. Dunn, P. Pearce, D. McCoy, G. M. Voelker, S. Savage, and K. Levchenko, "Reading the tea leaves: A comparative analysis of threat intelligence," in *Proc. 28th Secur. Symp. (Security 19)*, May 2019, pp. 851–867. Accessed: May 06, 2023. [Online]. Available: https://www.usenix.org/conference/usenixsecurity19/presentation/li
[30] F. Maymí, R. Bixler, R. Jones, and S. Lathrop, "Towards a definition of cyberspace tactics, techniques and procedures," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2017, pp. 4674–4679, doi: 10.1109/BIGDATA.2017.8258514.
[31] Y. He, L. Maglaras, A. Aliyu, and C. Luo, "Healthcare security incident response strategy–A proactive incident response (IR) procedure," *Secur. Commun. Netw.*, vol. 2022, pp. 1–10, Feb. 2022, doi: 10.1155/2022/2775249.
[32] E. Doynikova, E. Novikova, and I. Kotenko, "Attacker behaviour forecasting using methods of intelligent data analysis: A comparative review and prospects," *Information*, vol. 11, no. 3, p. 168, Mar. 2020, doi: 10.3390/info11030168.
[33] L. Qiang, Y. Zeming, L. Baoxu, J. Zhengwei, and Y. Jian, "Framework of cyber attack attribution based on threat intelligence," in *Proc. Int. Conf. Safety Secur. IoT*, in Lecture Notes of Institute Computer Sciences, Social-Informatics Telecommunications Engineering, vol. 190, 2017, pp. 92–103, doi: 10.1007/978-3-319-52727-7_11.
[34] *The NIST Cybersecurity Framework (CSF)*, NIST, Gaithersburg, MD, USA, Feb. 2024, doi: 10.6028/NIST.CSWP.29.
[35] Cambridge Dictionary. *PROACTIVE | English Meaning-Cambridge Dictionary*. Accessed: Aug. 14, 2024. [Online]. Available: https://dictionary.cambridge.org/dictionary/english/proactive
[36] D. S. Kuraku, D. Kalla, F. Samaah, and N. Smith, "Cultivating proactive cybersecurity culture among IT professional to combat evolving threats," *Int. J. Electr., Electron. Comput.*, vol. 8, no. 6, pp. 01–07, 2023, doi: 10.22161/eec.86.1.
[37] R. L. Stokes, K. Kiernan, and J. Rollins. (2013). *Employing the Intelligence Cycle Process Model Within the Homeland Security Enterprise*. Defense technical information center. Accessed: Jun. 28, 2023. [Online]. Available: https://apps.dtic.mil/sti/citations/ADA621345
[38] Y. Wu, "Strengthening intelligence education with information-processing and knowledge-organization competencies," *J. Strategic Secur.*, vol. 6, no. 3, pp. 10–24, Oct. 2013, doi: 10.5038/1944-0472.6.3.2.
[39] U.S. Army. (2013). *Joint Intelligence Joint Publication 2-0*. Accessed: Jul. 2, 2023. [Online]. Available: https://irp.fas.org/doddir/dod/jp2_0.pdf
[40] S. Ainslie, D. Thompson, S. Maynard, and A. Ahmad, "Cyber-threat intelligence for security decision-making: A review and research agenda for practice," *Comput. Secur.*, vol. 132, Sep. 2023, Art. no. 103352, doi: 10.1016/j.cose.2023.103352.
[41] Y. W. Hwang, I. Y. Lee, H. Kim, H. Lee, and D. Kim, "Current status and security trend of OSINT," Dept. Softw. Converg., Soonchunhyang Univ., Asan, Republic of Korea, 2022, vol. 2022, p. 14, doi: 10.1155/2022/1290129.
[42] J. Pastor-Galindo, P. Nespoli, F. Gomez Marmol, and G. Martinez Perez, "The not yet exploited goldmine of OSINT: Opportunities, open challenges and future trends," *IEEE Access*, vol. 8, pp. 10282–10304, 2020, doi: 10.1109/ACCESS.2020.2965257.
[43] H. Gibson, "Acquisition and preparation of data for OSINT investigations," in *Open Source Intelligence Investigation* (Advanced Sciences and Technologies for Security Applications), 2016, pp. 69–93, doi: 10.1007/978-3-319-47671-1_6.
[44] S. Rahman, "The art of open source intelligence (OSINT): Addressing cybercrime, opportunities, and challenges," *SSRN*, p. 30, Jun. 2025, doi: 10.2139/SSRN.5281845.
[45] M. Schäfer, M. Fuchs, M. Strohmeier, M. Engel, M. Liechti, and V. Lenders, "BlackWidow: Monitoring the dark web for cyber security information," in *Proc. 11th Int. Conf. Cyber Conflict (CyCon)*, vol. 900, May 2019, pp. 1–21, doi: 10.23919/CYCON.2019.8756845.
[46] M. K. Bergman, "White paper: The deep web: Surfacing hidden value," *J. Electron. Publishing*, vol. 7, no. 1, p. 17, Aug. 2001, doi: 10.3998/3336451.0007.104.
[47] S. Szymoniak and K. Foks, "Open source intelligence opportunities and challenges: A review," *Adv. Sci. Technol. Res. J.*, vol. 18, no. 3, pp. 123–139, 2024, doi: 10.12913/22998624/186036.
[48] R. Gill. *Sock Puppets in OSINT | SANS Institute*. Accessed: Jul. 19, 2023. [Online]. Available: https://www.sans.org/blog/what-are-sock-puppets-in-osint/
[49] D. T. Miller and J. S. Nye. (Nov. 2015). *Defense 2045: Assessing the Future Security Environment and Implications for Defense Policymakers*. CSIS Int. Secur. Program. Accessed: Jul. 9, 2023. [Online]. Available: https://espas.secure.europarl.europa.eu/orbis/sites/default/files/generated/document/en/151106_Miller_Defense2045_Web.pdf

[50] G. Yurtalan and S. Arslan, "Redefining OSINT software architecture with system-centric architecture design: A framework shaped by QAW, ADD, and ATAM," *IEEE Access*, vol. 13, pp. 71456–71480, 2025, doi: 10.1109/ACCESS.2025.3562718.

[51] S. Shafee, A. Bessani, and P. M. Ferreira, "Evaluation of LLM-based chatbots for OSINT-based cyber threat awareness," *Expert Syst. Appl.*, vol. 261, Feb. 2025, Art. no. 125509, doi: 10.1016/j.eswa.2024.125509.

[52] OSINT Community. *OSINT Framework*. Accessed: Jul. 14, 2023. [Online]. Available: https://osintframework.com/

[53] A. M. Sermakani, P. Sreejith, A. L. Krishna, S. Lokesh, C. V. R. K. Narayanan, and N. J. S. Subramaniam, "OPEN EYE: An information gathering tool using OSINT framework," in *Proc. Int. Conf. Smart Technol. Sustain. Develop. Goals (ICSTSDG)*, Jun. 2025, pp. 1–6, doi: 10.1109/ICSTSDG61998.2024.11026654.

[54] S. Bromander, M. Swimmer, L. P. M'uller, A. Jøsang, M. Eian, G. Skjøtskift, and F. Borg, "Investigating sharing of cyber threat intelligence and proposing a new data model for enabling automation in knowledge representation and exchange," *Digit. Threats, Res. Pract. (DTRAP)*, vol. 3, no. 1, pp. 1–22, Oct. 2021, doi: 10.1145/3458027.

[55] C. Sauerwein, C. Sillaber, A. Mussmann, and R. Breu, "Threat intelligence sharing platforms: An exploratory study of software vendors and research perspectives," in *Proc. Wirtschaftsinformatik*, 2017, pp. 837–851. [Online]. Available: https://aisel.aisnet.org/wi2017/track08/paper/3

[56] R. Azevedo, I. Medeiros, A. N. Bessani, and A. Bessani. (2018). *Automated Solution for Enrichment and Quality IoC Creation From OSINT*. [Online]. Available: https://www.researchgate.net/publication/327835294

[57] A. Tundis, S. Ruppert, and M. Mühlhäuser, "A feature-driven method for automating the assessment of OSINT cyber threat sources," *Comput. Secur.*, vol. 113, Feb. 2022, Art. no. 102576, doi: 10.1016/j.cose.2021.102576.

[58] P. M. Maniscalco, C. P. Holstege, and S. B. Cormier, "Operations security, site security, and incident response," in *Ciottone's Disaster Medicine*. Amsterdam, The Netherlands: Elsevier, 2024, doi: 10.1016/b978-0-323-80932-0.00091-4.

[59] P. Gao, F. Shao, X. Liu, X. Xiao, Z. Qin, F. Xu, P. Mittal, S. R. Kulkarni, and D. Song, "Enabling efficient cyber threat hunting with cyber threat intelligence," in *Proc. IEEE 37th Int. Conf. Data Eng. (ICDE)*, Apr. 2021, pp. 193–204, doi: 10.1109/ICDE51399.2021.00024.

[60] D. Gunter and M. Seitz. (2021). *SANS-A Practical Model for Conducting Cyber Threat Hunting*. Accessed: Jul. 19, 2023. [Online]. Available: https://www.sans.org/white-papers/38710/

[61] A. O. Bryushinin, A. V. Dushkin, and M. A. Melshiyan, "Automation of the information collection process by osint methods for penetration testing during information security audit," in *Proc. Conf. Russian Young Res. Electr. Electron. Eng. (ElConRus)*, Jan. 2022, pp. 242–246, doi: 10.1109/ElConRus54750.2022.9755812.

[62] *Framework for Improving Critical Infrastructure Cybersecurity*, NIST, Gaithersburg, MD, USA, Apr. 2018, doi: 10.6028/NIST.CSWP.04162018.

[63] P. Nespoli, D. Papamartzivanos, F. Gomez Marmol, and G. Kambourakis, "Optimal countermeasures selection against cyber attacks: A comprehensive survey on reaction frameworks," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1361–1396, 2nd Quart., 2018, doi: 10.1109/COMST.2017.2781126.

[64] P. An, R. Shafi, T. Mughogho, and O. Allan Onyango, "Multilingual email phishing attacks detection using OSINT and machine learning," 2025, *arXiv:2501.08723*.

[65] Merriam-Webster. *Public Sector Definition & Meaning*. Accessed: Dec. 2, 2024. [Online]. Available: https://www.merriam-webster.com/dictionary/public%20sector

[66] H. L. Larsen, J. M. Blanco, R. P. Pastor, and R. R. Yager, *Using Open Data to Detect Organized Crime Threats: Factors Driving Future Crime*. Germany: Springer, May 2017, pp. 1–282, doi: 10.1007/978-3-319-52703-1.

[67] D. Van Puyvelde and F. Tabárez Rienzi, "The rise of open-source intelligence," *Eur. J. Int. Secur.*, pp. 1–15, Jan. 2025, doi: 10.1017/eis.2024.61.

[68] J. Zhou, "Open-source intelligence and great-power competition under mediatization," *Secur. J.*, vol. 37, no. 4, pp. 1769–1786, Dec. 2024, doi: 10.1057/s41284-024-00446-0.

[69] A. Mackinnon. (2020). *How Bellingcat's Open Source Investigations Help the Work of U.S. Intelligence Agencies*. Accessed: Jul. 18, 2023. [Online]. Available: https://foreignpolicy.com/2020/12/17/bellingcat-can-say-what-u-s-intelligence-cant/

[70] Merriam-Webster. *Private Sector Definition & Meaning*. Accessed: Dec. 2, 2024. [Online]. Available: https://www.merriam-webster.com/dictionary/private%20sector

[71] World Bank Group. (2015). *IFC Jobs Study: Assessing Private Sector Contributions to Job Creation and Poverty Reduction*. Accessed: Dec. 2, 2024. [Online]. Available: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/1571914683 26714061/ifc-jobs-study-assessing-private-sector-contributions-to-job-creation-and-poverty-reduction

[72] UNDRR. (2019). *Roundtable on 'Finance for Disaster Risk Reduction and Boosting Public-Private Sector Cooperation' | UNDRR*. Accessed: Dec. 2, 2024. [Online]. Available: https://www.undrr.org/news/roundtable-finance-disaster-risk-reduction-and-boosting-public-private-sector-cooperation

[73] DNI-USA. (2009). *An Overview of the United States Intelligence Community for the 111th Congress*. Accessed: Jul. 23, 2023. [Online]. Available: https://apps.dtic.mil/sti/citations/ADA493616

[74] C. Weinbaum, B. M. Knopp, S. Kim, and Y. Shokh, "Options for strengthening all-source intelligence: Substantive change is within reach," RAND Nat. Secur. Res. Division (NSRD), CA, USA, Tech. Rep., Feb. 2022, doi: 10.7249/RRA1245-1.

[75] B. Krilov. (2022). *Ethical Frameworks in Open-Source Intelligence*. Accessed: Jun. 17, 2023. [Online]. Available: https://www.dhs.gov/sites/default/files/2022-09/Ethical%20Frameworks%20in%20OSINT%20 Final.pdf

[76] P. T. Jaeger, J. C. Bertot, and C. R. McClure, "The impact of the USA patriot act on collection and analysis of personal information under the foreign intelligence surveillance act," *Government Inf. Quart.*, vol. 20, no. 3, pp. 295–314, Jul. 2003, doi: 10.1016/s0740-624x(03)00057-1.

[77] K. M. Vogel, "OSINT and the U.S. intelligence community: Is the past prologue?" in *Open Source Investigations in the Age of Google*, 2024, pp. 188–203, doi: 10.1142/9781800614079_0010.

[78] M. G. Arvanitidou and S. Prokesch. (2020). *Deep Learning in Multimedia*. FKT-Fernseh-und Kinotechnik. Accessed: Jul. 20, 2023. [Online]. Available: https://science.fkt-online.de/articles/34/galley/34/download/

[79] T. C. Helmus. (2022). *Artificial Intelligence, Deepfakes, and Disinformation: A Primer*. Accessed: Jul. 20, 2023. [Online]. Available: https://www.jstor.org/stable/resrep42027

[80] A. Powell and C. Haynes, "Social media data in digital forensics investigations," in *Digital Forensic Education*, vol. 61. Cham, Switzerland: Springer, 2020, doi: 10.1007/978-3-030-23547-5_14.

[81] E. Nicart, B. Zanuttini, H. Gilbert, B. Grilheres, and F. Praca, "Building document treatment chains using reinforcement learning and intuitive feedback," in *Proc. IEEE 28th Int. Conf. Tools Artif. Intell. (ICTAI)*, Nov. 2016, pp. 635–639, doi: 10.1109/ICTAI.2016.0102.

[82] A. Gandomi and M. Haider, "Beyond the hype: Big data concepts, methods, and analytics," *Int. J. Inf. Manage.*, vol. 35, no. 2, pp. 137–144, Apr. 2015, doi: 10.1016/j.ijinfomgt.2014.10.007.

[83] A. Barnea, "Big data and counterintelligence in western countries," *Int. J. Intell. CounterIntelligence*, vol. 32, no. 3, pp. 433–447, Jul. 2019, doi: 10.1080/08850607.2019.1605804.

[84] T. Day, H. Gibson, and S. Ramwell, "Fusion of OSINT and Non-OSINT data," in *Open Source Intelligence Investigation* (Advanced Sciences and Technologies for Security Applications). Germany: Springer, 2016, pp. 133–152, doi: 10.1007/978-3-319-47671-1_9.

[85] F. Tabatabaei and D. Wells, "OSINT in the context of cyber-security," in *Advanced Sciences and Technologies for Security Applications*, 2016, pp. 213–231.

[86] C. S. Fleisher, "Using open source data in developing competitive and marketing intelligence," *Eur. J. Marketing*, vol. 42, no. 7/8, pp. 852–866, Jul. 2008, doi: 10.1108/03090560810877196.

[87] G. Bello-Orgaz, J. J. Jung, and D. Camacho, "Social big data: Recent achievements and new challenges," *Inf. Fusion*, vol. 28, pp. 45–59, Mar. 2016, doi: 10.1016/j.inffus.2015.08.005.

[88] F. Gómez Mármol, M. Gil Pérez, and G. Martínez Pérez, "Reporting offensive content in social networks: Toward a reputation-based assessment approach," *IEEE Internet Comput.*, vol. 18, no. 2, pp. 32–40, Mar. 2014, doi: 10.1109/MIC.2013.132.

[89] S. Szymoniak, K. Foks, and A. Pyrkosz-Dziubczyk, "Application of OSINT methods in ensuring cybersecurity," *IPSI Trans. Internet Res.*, vol. Volume 21, no. Issue 2, p. 38, May 2025, doi: 10.58245/ipsi.tir.2502.05.

[90] S. Gong, J. Cho, and C. Lee, "A reliability comparison method for OSINT validity analysis," *IEEE Trans. Ind. Informat.*, vol. 14, no. 12, pp. 5428–5435, Dec. 2018, doi: 10.1109/TII.2018.2857213.

[91] M. Zago, P. Nespoli, D. Papamartzivanos, M. G. Perez, F. G. Marmol, G. Kambourakis, and G. M. Perez, "Screening out social bots interference: Are there any silver bullets?" *IEEE Commun. Mag.*, vol. 57, no. 8, pp. 98–104, Aug. 2019, doi: 10.1109/MCOM.2019.1800520.
[92] G. R. S. Weir, "The limitations of automating OSINT: Understanding the question, not the answer," in *Automating Open Source Intelligence*, 2016, pp. 159–169.
[93] P. Casanovas, "Cyber warfare and organised Crime. A regulatory model and meta-model for open source intelligence (OSINT)," in *Ethics and Policies for Cyber Operations*, 2017, pp. 139–167, doi: 10.1007/978-3-319-45300-2_9.
[94] H. Bean, "Is open source intelligence an ethical issue?" *Res. Social Problems Public Policy*, vol. 19, pp. 385–402, Jan. 2011, doi: 10.1108/S0196-1152(2011)0000019024.
[95] M. Kandias, L. Mitrou, V. Stavrou, and D. Gritzalis, "Which side are you on? A new panopticon vs. Privacy," in *Proc. Int. Conf. Secur. Cryptogr. (SECRYPT)*, Jul. 2013, pp. 1–13.
[96] J. Simola, "Privacy issues and critical infrastructure protection," in *Emerging Cyber Threats and Cognitive Vulnerabilities*, 2020, pp. 197–226.
[97] T. O. Browne, M. Abedin, and M. J. M. Chowdhury, "A systematic review on research utilising artificial intelligence for open source intelligence (OSINT) applications," *Int. J. Inf. Secur.*, vol. 23, no. 4, pp. 2911–2938, Aug. 2024, doi: 10.1007/s10207-024-00868-2.
[98] M. Lee, "Generating intelligence," in *Cyber Threat Intelligence*. New York, NY, USA: Wiley, Apr. 2023, pp. 127–154, doi: 10.1002/9781119861775.CH5.
[99] H. Zhang and Y. Lu, "Web 3.0: Applications, opportunities and challenges in the next internet generation," *Syst. Res. Behav. Sci.*, p. 20, Mar. 2025, doi: 10.1002/sres.3151.
[100] K. Gradoń, "Crime in the time of the plague: Fake news pandemic and the challenges to law-enforcement and intelligence community," *Soc. Register*, vol. 4, no. 2, pp. 133–148, Apr. 2020, doi: 10.14746/sr.2020.4.2.10.

ZAFRIR AVRAHAMI received the Electronic Engineering degree with a specialization in computer communication and the M.B.A. degree in entrepreneurship and future technologies from Ariel University, 2022. In 2019, he received the Chief Information Security Officer (CISO) Certification from The Hebrew University of Jerusalem. Additionally, he was a Cybersecurity Project Manager with Israel's National Cyber Directorate (INCD), focusing on national cyber defense. He is currently a Researcher with the Department of Business Administration, Ariel University, focusing on cybersecurity. He has approximately 15 years of professional experience as a technology project manager in the public sector, specializing in communication, internet, cybersecurity, and intelligence. His main research interests include cyber intelligence, the human factor in cybersecurity, and human error in cybersecurity.

MOTI ZWILLING received the Ph.D. degree in computer science and biology from The Hebrew University of Jerusalem, Israel, and the Ph.D. degree in business administration from Bar-Ilan University, Israel. He currently holds the position of an Associate Professor with Ariel University and a Research Fellow with the Ariel Cyber Innovation Center. Recognized worldwide as an expert in AI, cybersecurity, management, and data science, he collaborates with the Department of Biological Engineering, MIT, USA, on data science research. He has extensive experience in research and development and project management within the Israeli high-tech industry. Additionally, he serves on the advisory board and as an Associate Editor for several journals, including *Computer Information Systems*.

CHEN HAJAJ (Member, IEEE) received the B.Sc. degree in computer engineering, the M.Sc. degree in electrical engineering, and the Ph.D. degree in computer science from Bar-Ilan University.

He is currently a Faculty Member with the Department of Industrial Engineering and Management, Ariel University, and a Researcher with the Ariel Data Science and Artificial Intelligence Research Center. From 2016 to 2018, he was a Postdoctoral Fellow with Vanderbilt University. He has published extensively in leading journals and conferences in these areas, and his work has been supported by competitive research grants, including funding from the Ministry of Innovation, Science, and Technology. His research interests include the intersection of machine learning, game theory, and cybersecurity, with a particular emphasis on encrypted traffic classification, adversarial artificial intelligence, and multimodal learning.